Diary Data Analysis

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2025-08-12

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Exploratory: Impact of CU on training

1) Child rated ASB at 2 weeks - by CU

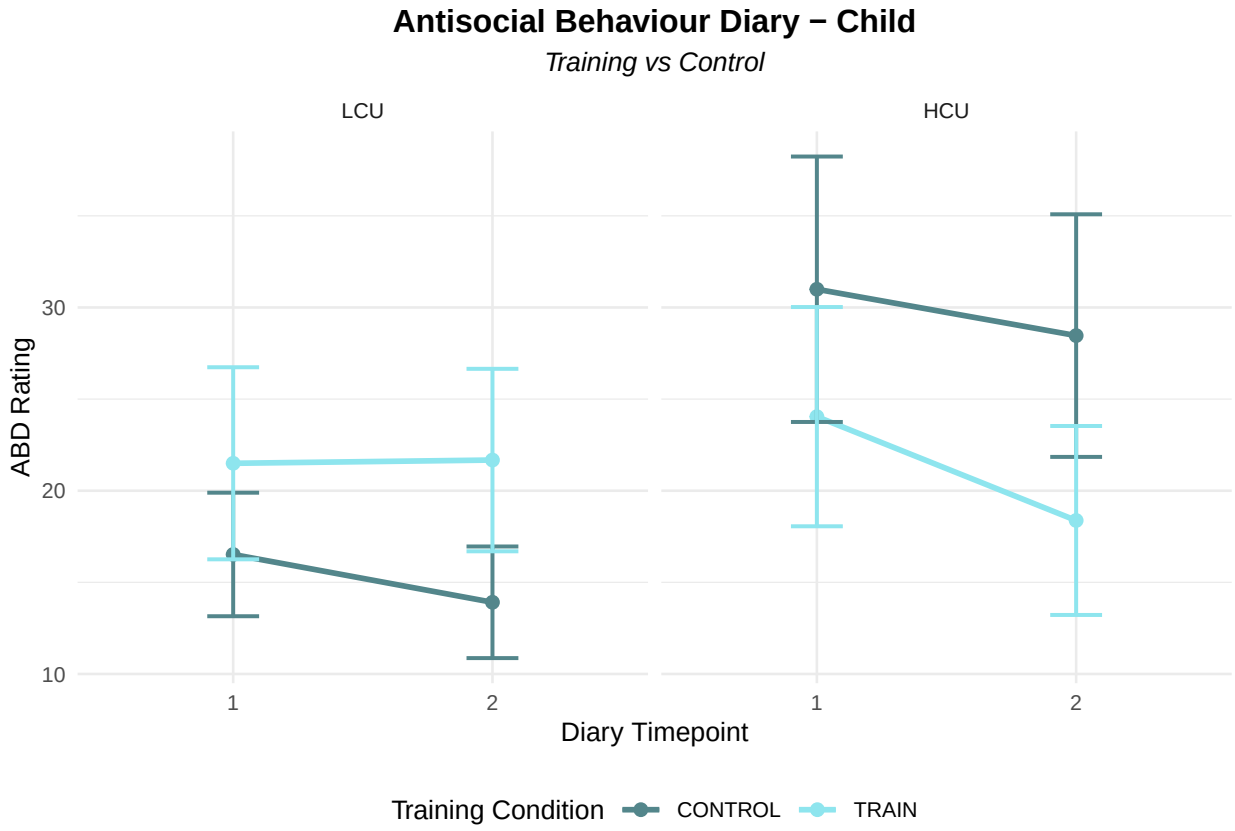
1.1) Table of outcomes

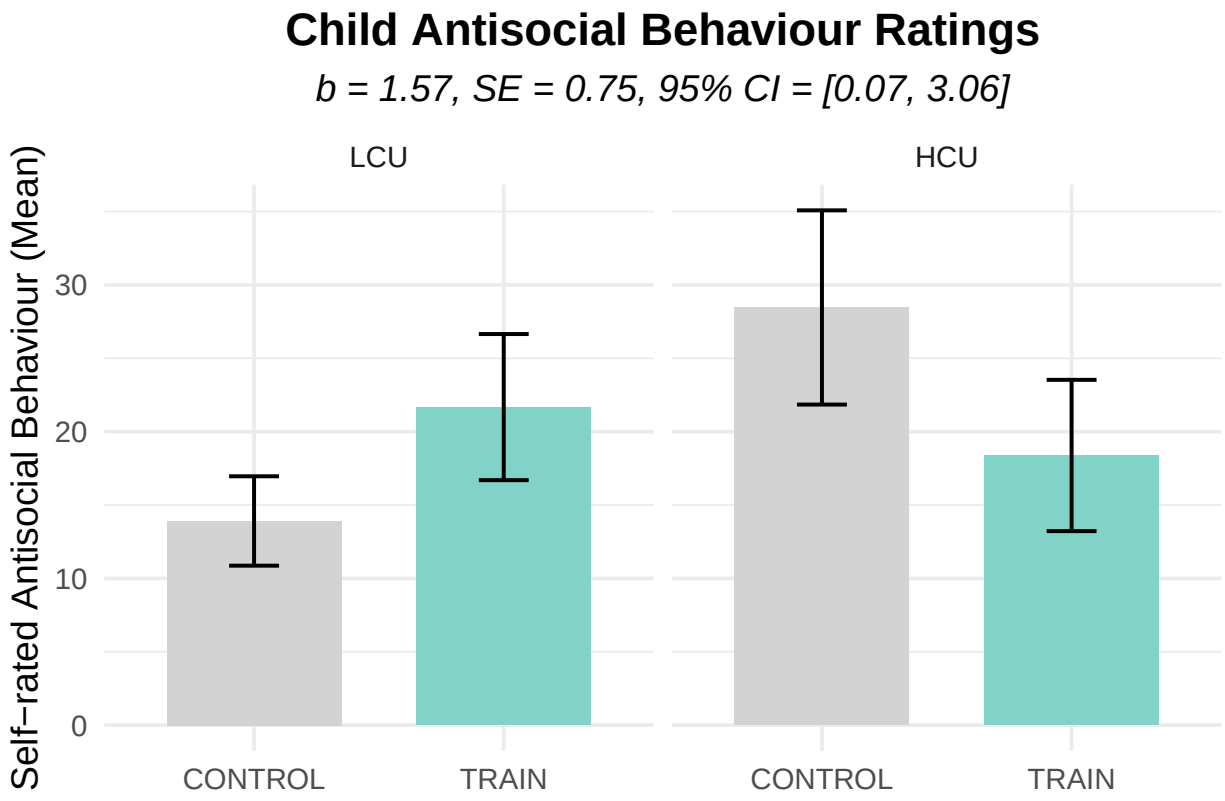
Table of outcomes:

Training_Condition	diaryTP	CP_Group	Group_N_ABD	Mean_child_rated_ABD	se_child_rated_ABD	NA_child_rated_ABD
CONTROL	1	LCU	26	16.52077	3.370843	0
CONTROL	1	HCU	18	30.99333	7.242612	0
CONTROL	2	LCU	26	13.91154	3.045668	0
CONTROL	2	HCU	18	28.46444	6.619444	0
TRAIN	1	LCU	22	21.49636	5.240413	0
TRAIN	1	HCU	16	24.04125	5.982017	0
TRAIN	2	LCU	22	21.67318	4.978893	0
TRAIN	2	HCU	16	18.37625	5.155261	0

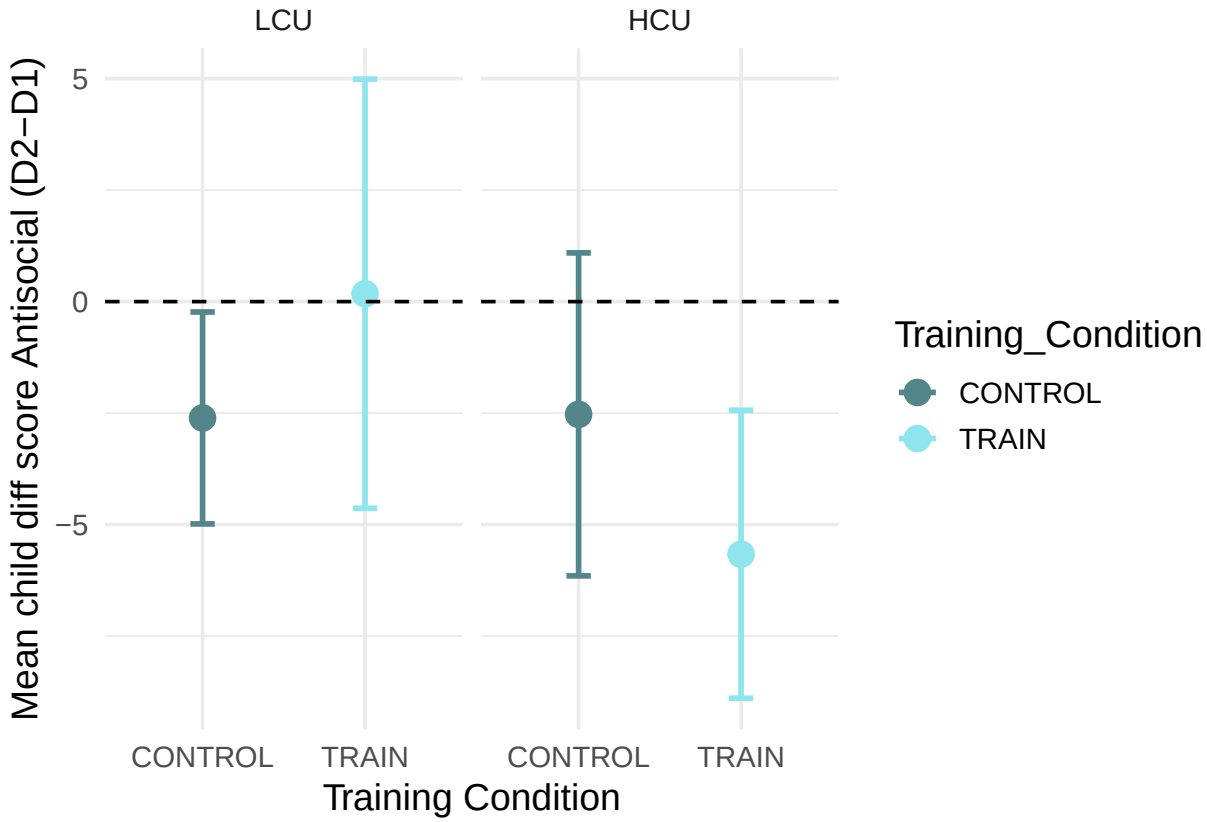
1.3) Plot of outcomes

First plotting data:




Child Antisocial Behaviour Ratings
b = 1.57, SE = 0.75, 95% CI = [0.07, 3.06]
LCU
HCU
Self−rated Antisocial Behaviour (Mean)
30
20
10
0
CONTROL
TRAIN
CONTROL
TRAIN

Difference plot:



1.3) Linear Regression models - in line with other analyses

```
# simple linear model (in line with other analyses)
model_c_asb_CU_lm <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + CP_Group +
                         Training_Condition*CP_Group,
                       data =  child_asb_d1d2_wide)

# model without interaction
model_c_asb_CU_lm_nointeraction <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + CP_Group,
                       data =  child_asb_d1d2_wide)
```

1.3.1) Does interaction add anything to model? - No

```
## Analysis of Variance Table
##
## Model 1: ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + CP_Group
## Model 2: ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + CP_Group + Training_Condition *
##     CP_Group
##   Res.Df   RSS Df Sum of Sq     F Pr(>F)
## 1     78 16854
## 2     77 16396  1    458.66 2.154 0.1463
```

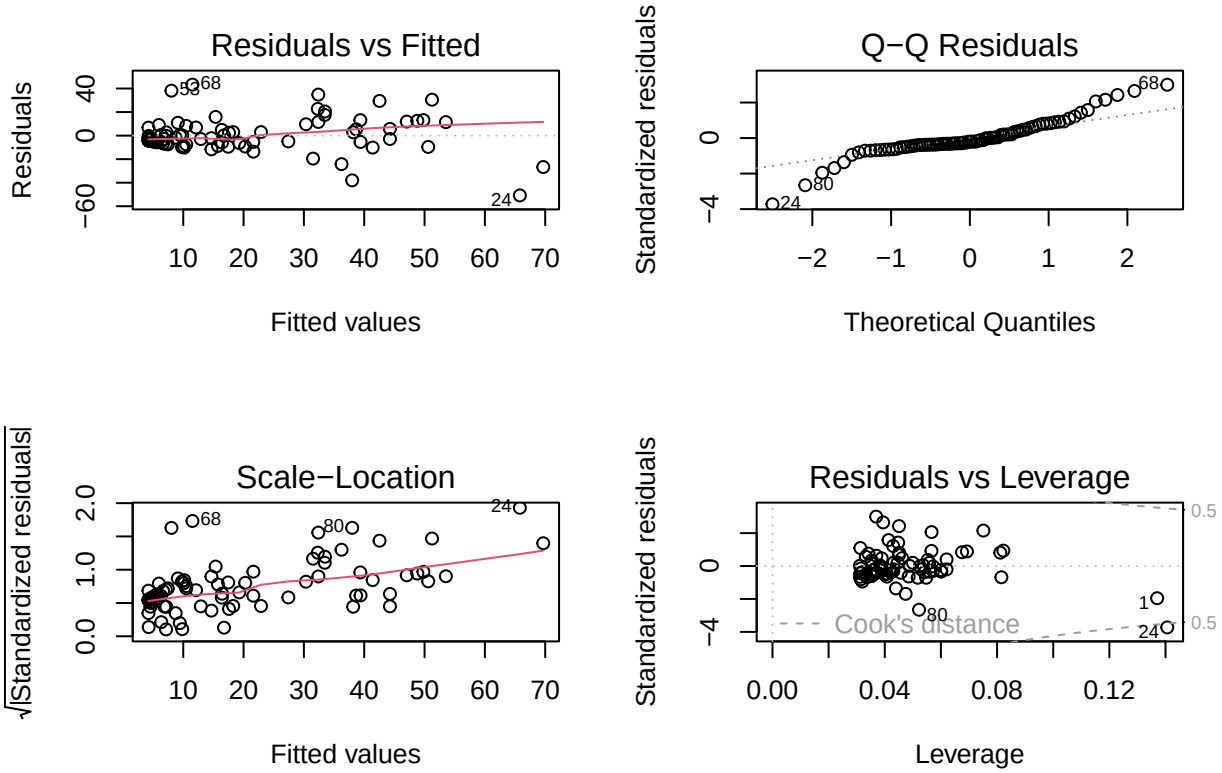
1.3.2) LM Results:

```
##
## Call:
## lm(formula = ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + CP_Group,
##     data = child_asb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -50.801  -5.818  -2.901   6.513  43.306
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)               4.23401    2.89971   1.460    0.148
## ASB_sum_D1                0.69586    0.06898  10.088 8.53e-16 ***
## Training_ConditionTRAIN   0.33151    3.25558   0.102    0.919
## CP_GroupHCU               0.03662    3.35219   0.011    0.991
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.7 on 78 degrees of freedom
## Multiple R-squared:  0.5747, Adjusted R-squared:  0.5583
## F-statistic: 35.13 on 3 and 78 DF,  p-value: 1.805e-14


##
## Call:
## lm(formula = ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition + CP_Group +
##     Training_Condition * CP_Group, data = child_asb_d1d2_wide)
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -52.111  -7.113  -2.481   7.771  41.012
##
## Coefficients:
##                                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)                         2.62381    3.08048   0.852    0.397
## ASB_sum_D1                          0.68325    0.06901   9.901 2.23e-15 ***
## Training_ConditionTRAIN             4.36209    4.24101   1.029    0.307
## CP_GroupHCU                         4.66460    4.58438   1.017    0.312
## Training_ConditionTRAIN:CP_GroupHCU -9.70031    6.60934  -1.468    0.146
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 14.59 on 77 degrees of freedom
## Multiple R-squared:  0.5862, Adjusted R-squared:  0.5647
## F-statistic: 27.27 on 4 and 77 DF,  p-value: 4.143e-14
```

1.3.3) LM Assum ption Checks



1.4) Bootstrap linear regression model (in line with other analyses)

1.4.1) Results and specification

```
boot_coef_c_asb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(ASB_sum_D2 ~ ASB_sum_D1 + Training_Condition*CP_Group,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
## Boot function: Generates R bootstrap replicates of a statistic applied to data.
## data = my data
## statistic = A function which, when applied to data returns a vector containing the statistic(s) of i
set.seed(123)  # For reproducibility
boot_results_c_asb <- boot(data = child_asb_d1d2_wide,
                     statistic = boot_coef_c_asb,
                     R = 5000)

# View results
boot_results_c_asb
```

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_asb_d1d2_wide, statistic = boot_coef_c_asb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##      original       bias    std. error
## t1*  2.623805 -0.134053368   2.5066037
## t2*  0.683245  0.009755394   0.1066876
## t3*  4.362093  0.197938460   4.5370893
## t4*  4.664599 -0.137212246   3.9131004
## t5* -9.700311 -0.145731359   6.1626729
```

1.4.2) Confidence intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-1.750,  8.241 )
## Calculations and Intervals on Original Scale
```

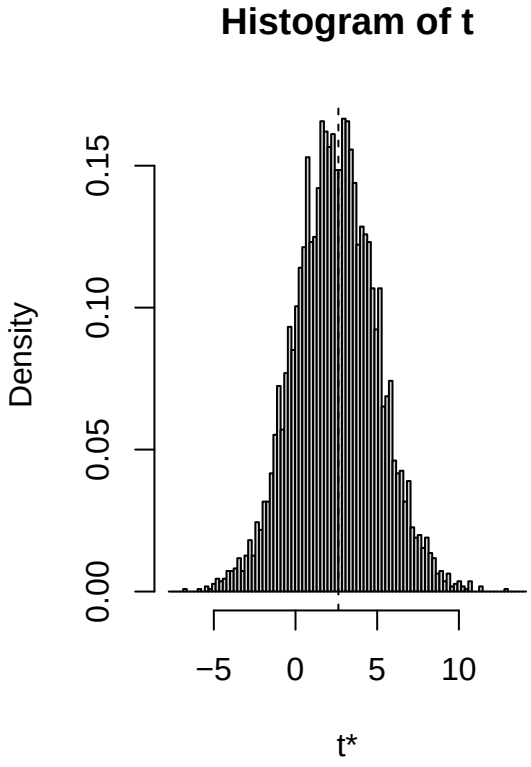
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.4520,  0.8715 )
## Calculations and Intervals on Original Scale

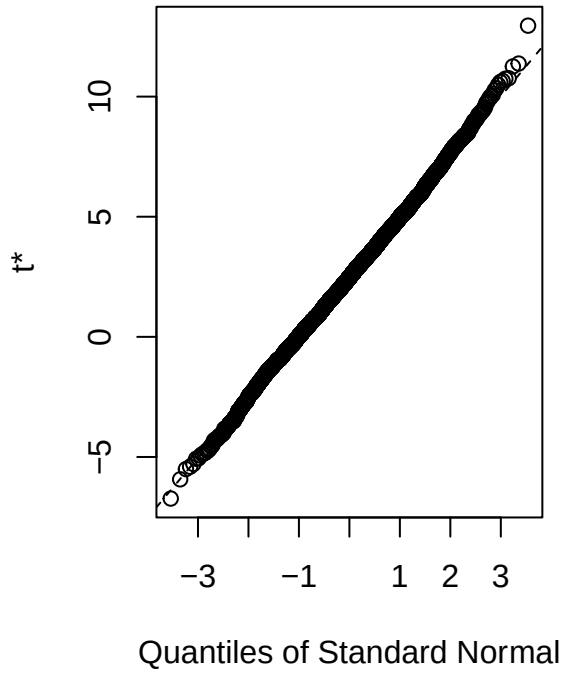
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-4.331, 13.638 )
## Calculations and Intervals on Original Scale

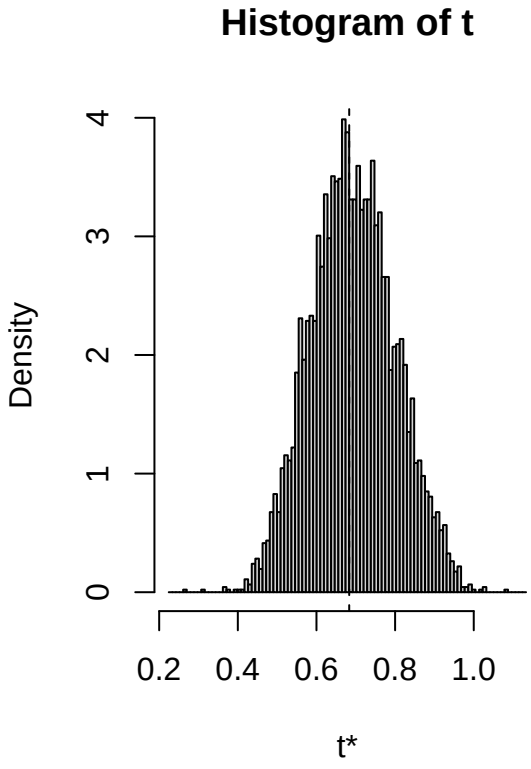
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-2.143, 13.324 )
## Calculations and Intervals on Original Scale

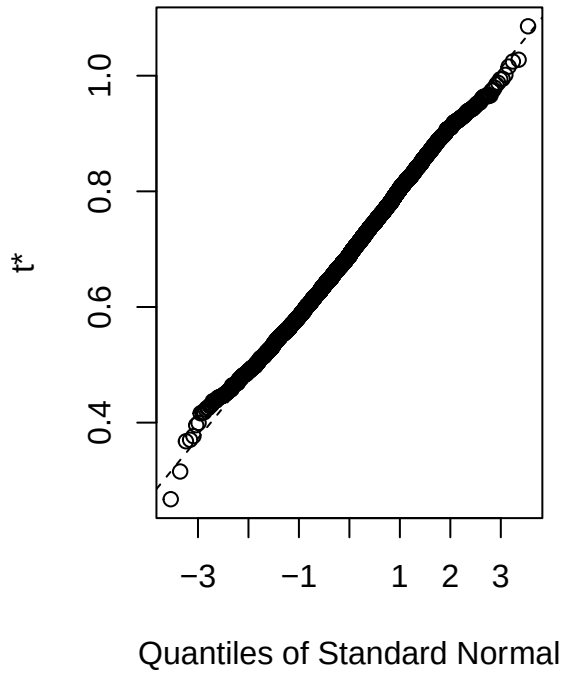
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_asb, type = "bca", index = 5)
##
## Intervals :
## Level       BCa
## 95%   (-23.326,   1.685 )
## Calculations and Intervals on Original Scale
```

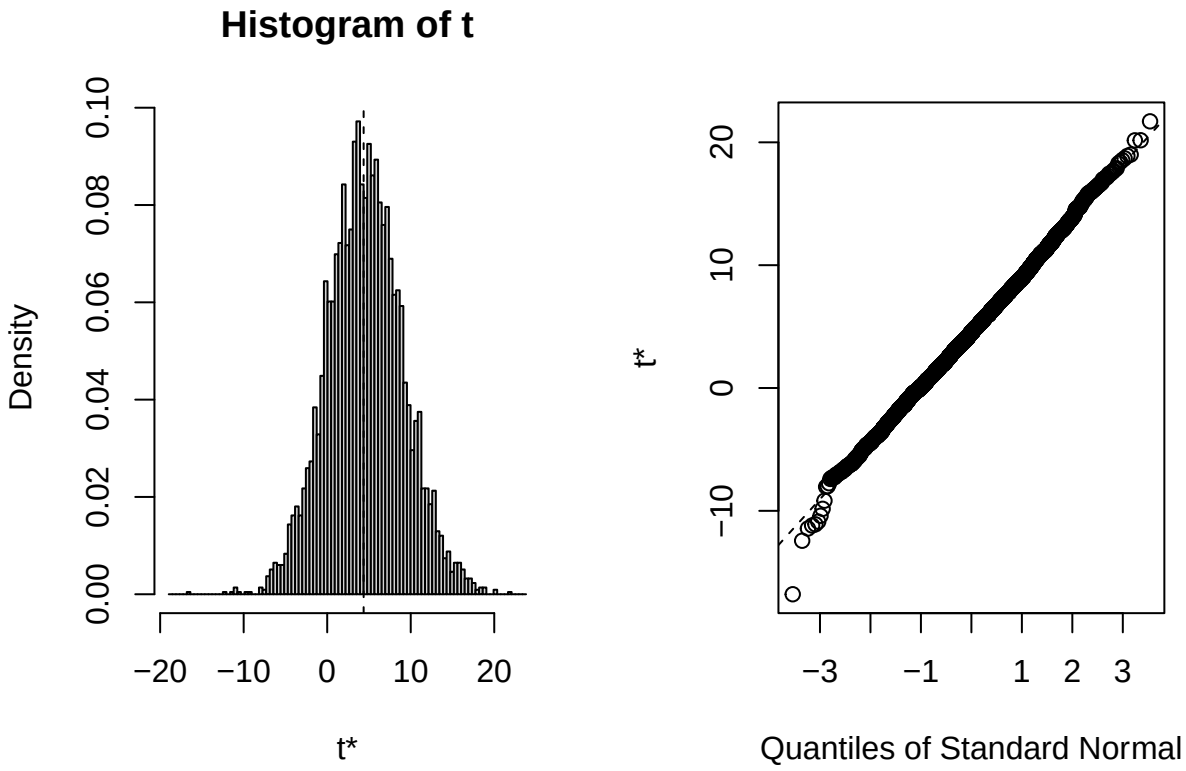
1.4.3) Model diagnostics

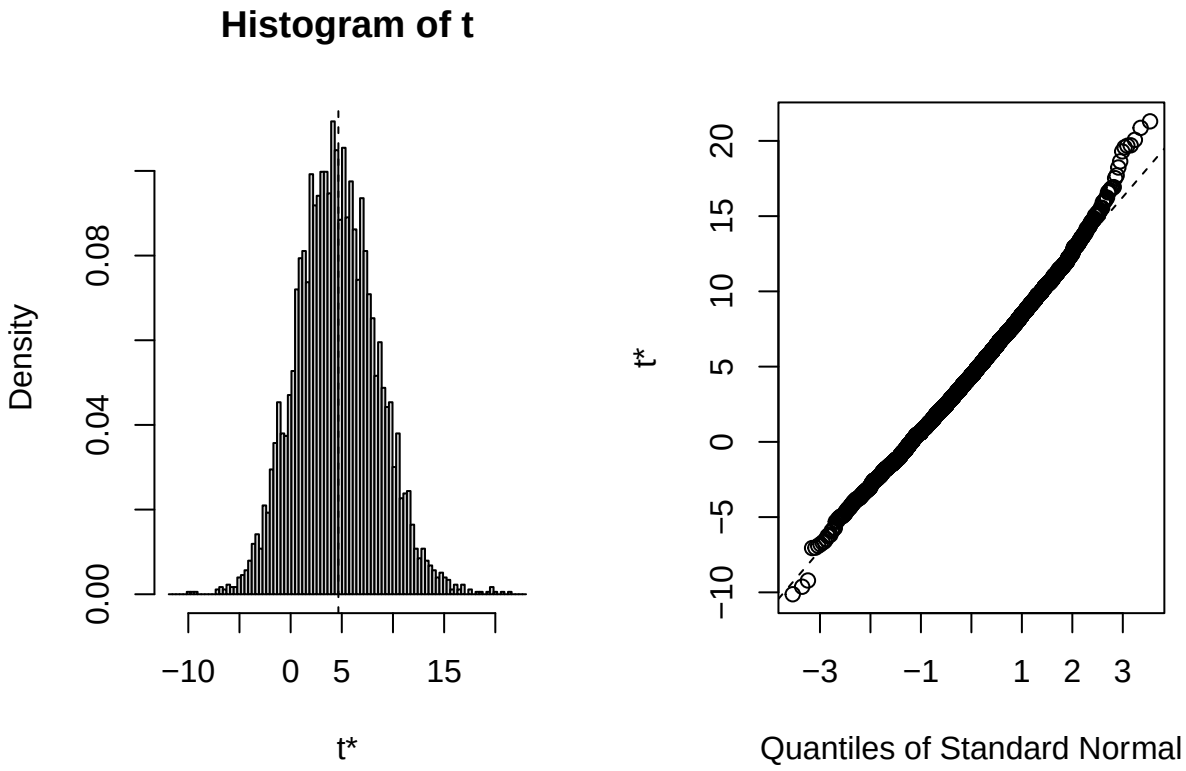


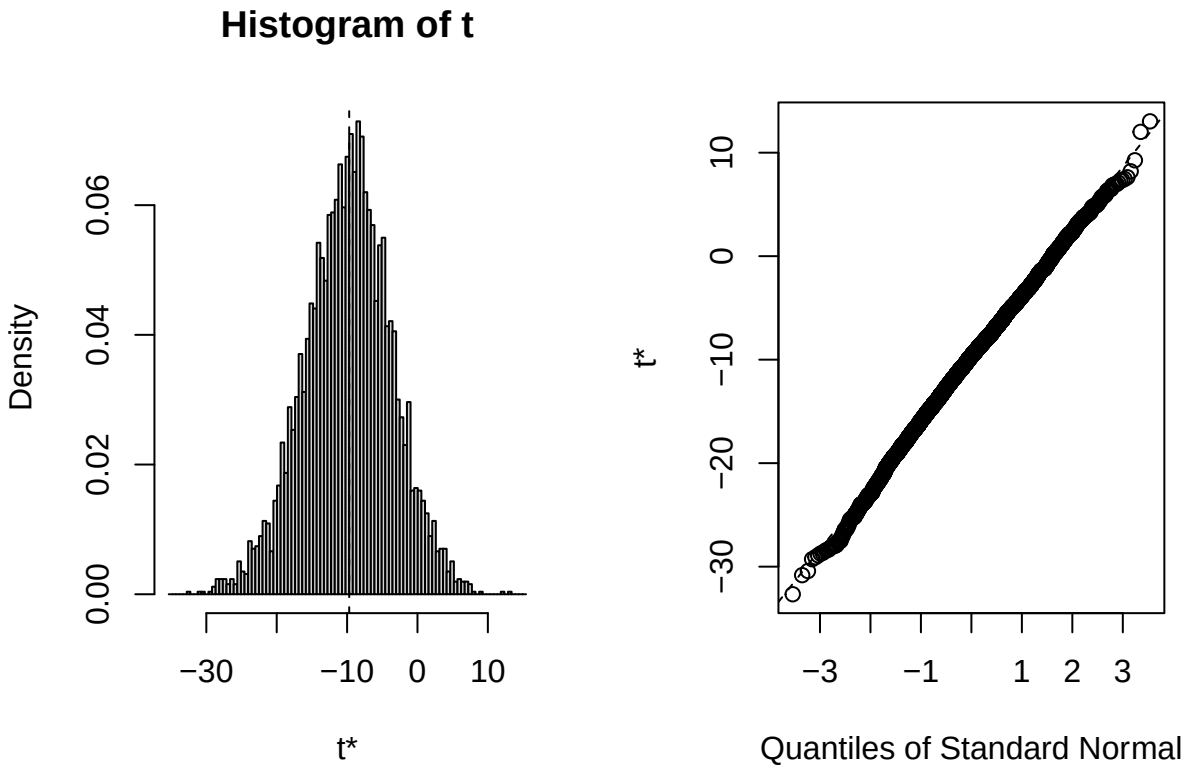







Histogram of t
Density
0.00
0.02
0.04
0.06
0.08
0.10
−20
−10
0
10
20
t*
t*
−10
0
10
20
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.04
0.08
−10
0
5
15
t*
t*
−10
−5
0
5
10
15
20
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.02
0.04
0.06
−30
−10
0
10
t*
t*
10
0
−10
−20
−30
−3
−1
1
2
3
Quantiles of Standard Normal

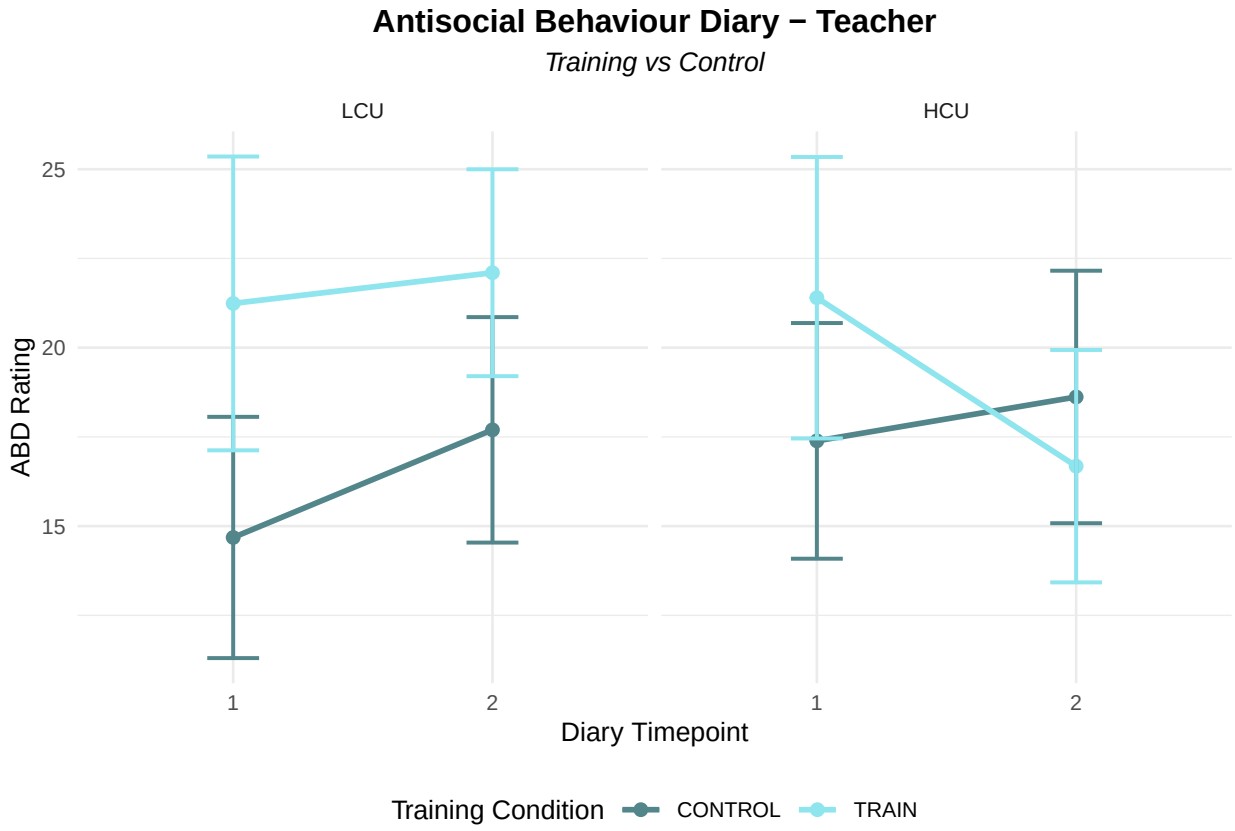
2) Teacher rated ASB at two weeks

2.1) Table of outcomes

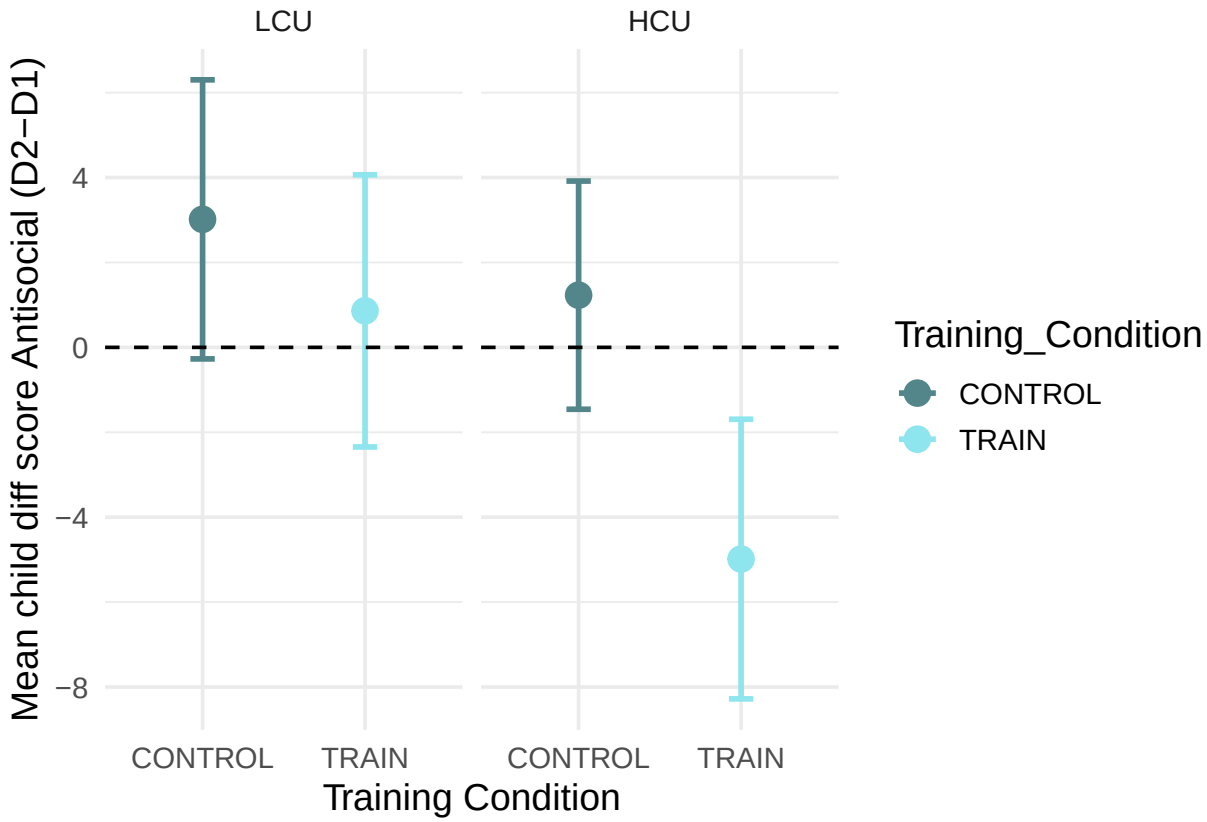
Training_Condition	diaryTP	CP_Group	Group_N	Mean_ABD	se_ABD	Mean_PBD	se_PBD
CONTROL	1	LCU	32	14.68229	3.379629	37.48958	4.023299
CONTROL	1	HCU	21	17.38889	3.301882	18.07937	3.305231
CONTROL	2	LCU	32	17.69792	3.158693	35.24479	3.616988
CONTROL	2	HCU	21	18.61905	3.537556	24.12698	4.220790
TRAIN	1	LCU	25	21.24000	4.115801	39.55333	4.566927
TRAIN	1	HCU	25	21.40000	3.943220	27.64000	3.650338
TRAIN	2	LCU	25	22.10000	2.898211	38.41333	4.598401
TRAIN	2	HCU	25	16.68000	3.256161	30.14667	3.809410

2.2) plots of outcomes

First plotting data:



Difference plot:


LCU
HCU
Mean child diff score Antisocial (D2−D1)
4
0
−4
−8
CONTROL
TRAIN
CONTROL
TRAIN
Training Condition
Training_Condition
CONTROL
TRAIN

2.3) Linear Regression model

```
lm_model_t_asb_CU <- lm(ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition + CP_Group +
                         Training_Condition*CP_Group,
                       data =  teach_asb_d1d2_wide)


# model without interaction
lm_model_t_asb_CU_nointeraction <- lm(ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition + CP_Group,
                       data =  teach_asb_d1d2_wide)
```

2.3.1) Is model with interaction better than model without? - No

```
## Analysis of Variance Table
##
## Model 1: ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition + CP_Group + Training_Condition *
##     CP_Group
## Model 2: ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition + CP_Group
##   Res.Df   RSS Df Sum of Sq     F Pr(>F)
## 1     97 17378
## 2     98 17526 -1   -148.16 0.827 0.3654
```

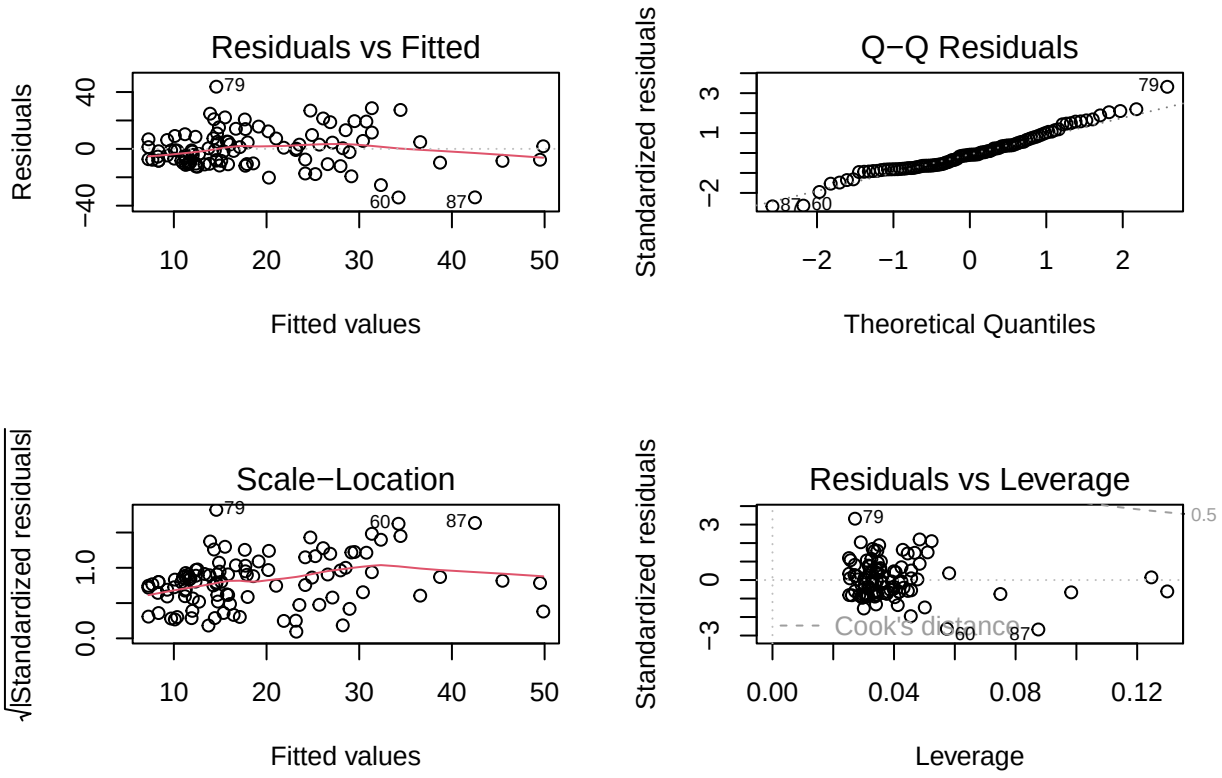
2.3.2) Model results

```
##
## Call:
## lm(formula = ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition +
##     CP_Group, data = teach_asb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -34.238  -8.983  -1.141   7.159  43.754
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              11.19668    2.37760   4.709 8.20e-06 ***
## ASB_mean_D1               0.50744    0.07154   7.093 2.06e-10 ***
## Training_ConditionTRAIN -1.08956    2.69451  -0.404    0.687
## CP_GroupHCU             -2.84772    2.68209  -1.062    0.291
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.37 on 98 degrees of freedom
## Multiple R-squared:  0.3432, Adjusted R-squared:  0.3231
## F-statistic: 17.07 on 3 and 98 DF,  p-value: 5.372e-09
```

```
##
## Call:
## lm(formula = ASB_mean_D2 ~ ASB_mean_D1 + Training_Condition +
##     CP_Group + Training_Condition * CP_Group, data = teach_asb_d1d2_wide)
##
```

```
## Residuals:
##     Min      1Q  Median      3Q     Max
## -33.139  -9.343  -2.253   6.266  44.691
##
## Coefficients:
##                                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)                        10.26868    2.58927   3.966  0.00014 ***
## ASB_mean_D1                         0.50600    0.07162   7.065 2.45e-10 ***
## Training_ConditionTRAIN             1.08388    3.60350   0.301  0.76422
## CP_GroupHCU                        -0.44841    3.76393  -0.119  0.90542
## Training_ConditionTRAIN:CP_GroupHCU -4.87818    5.36418  -0.909  0.36539
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.38 on 97 degrees of freedom
## Multiple R-squared:  0.3488, Adjusted R-squared:  0.3219
## F-statistic: 12.99 on 4 and 97 DF,  p-value: 1.656e-08
```

2.3.3) Assumption checks



2.4) Bootstrapped linear regression

2.4.1) Results and specification

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_asb_d1d2_wide, statistic = boot_coef_t_asb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##       original       bias    std. error
## t1* 10.2686789  0.018050068   2.6778729
## t2*  0.5059999  0.004261537   0.0863913
## t3*  1.0838837 -0.099274282   3.8012049
## t4* -0.4484069 -0.087626996   3.8543236
## t5* -4.8781807  0.100646481   5.2237335
```

2.4.2) Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   ( 5.94, 16.90 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.3328,  0.6741 )
## Calculations and Intervals on Original Scale

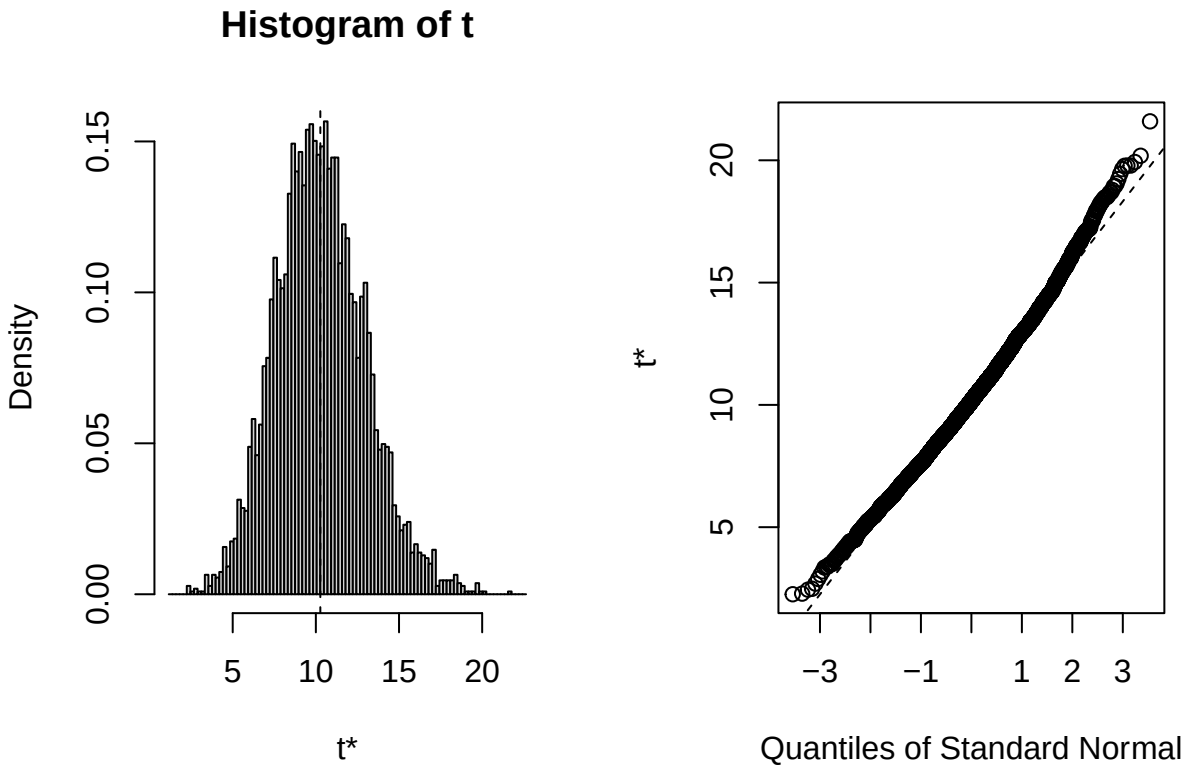
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
```

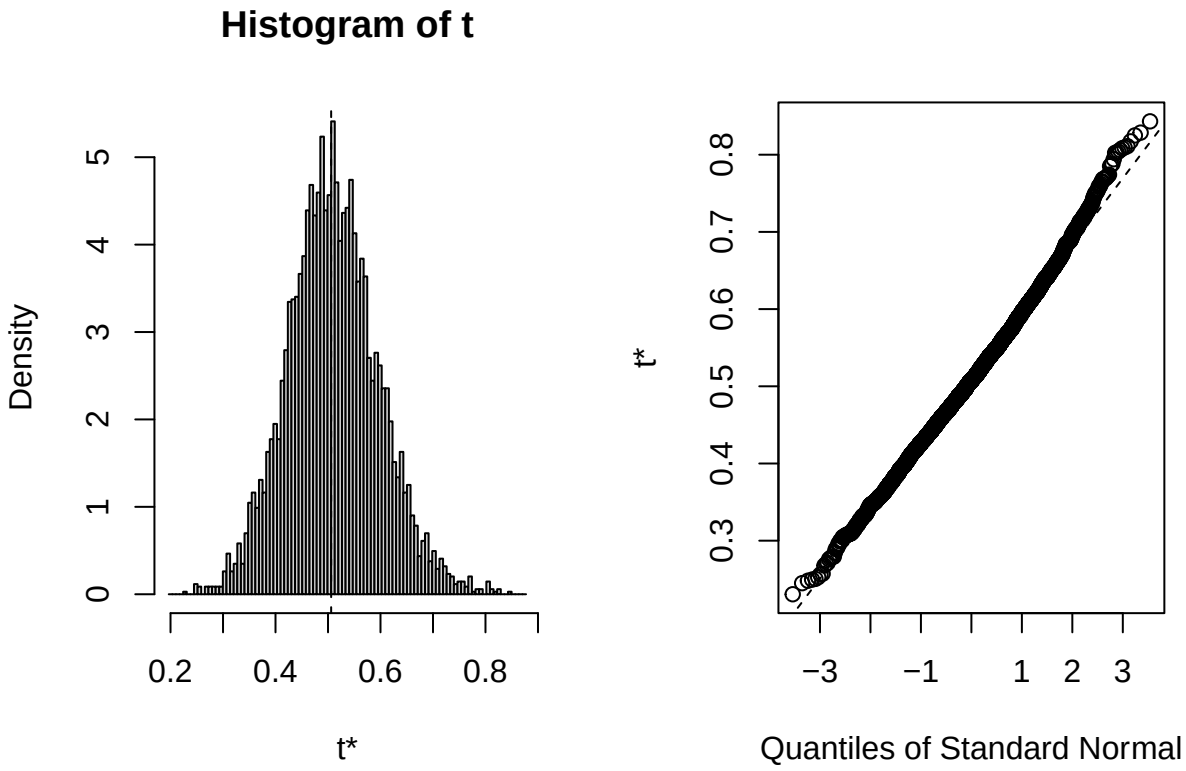
```
## 95%   (-6.504,  8.567 )
## Calculations and Intervals on Original Scale

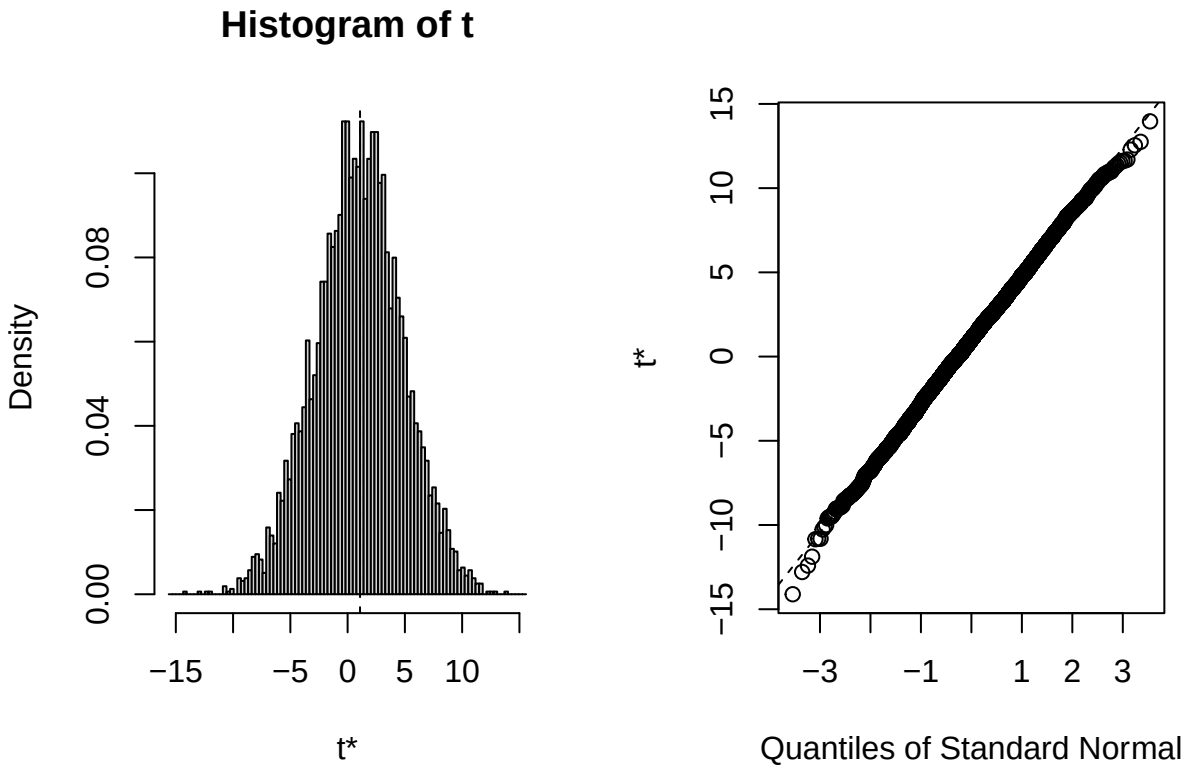
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-7.9679,  7.3238 )
## Calculations and Intervals on Original Scale

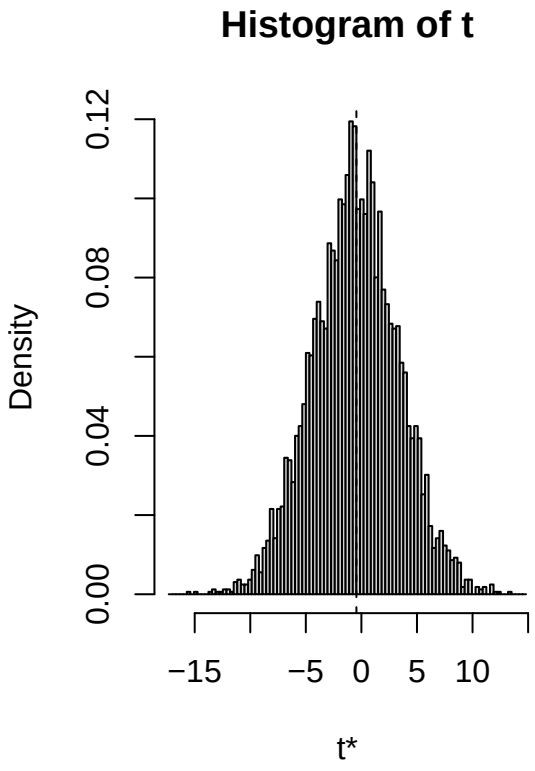
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_asb, type = "bca", index = 5)
##
## Intervals :
## Level       BCa
## 95%   (-15.564,   5.359 )
## Calculations and Intervals on Original Scale
```

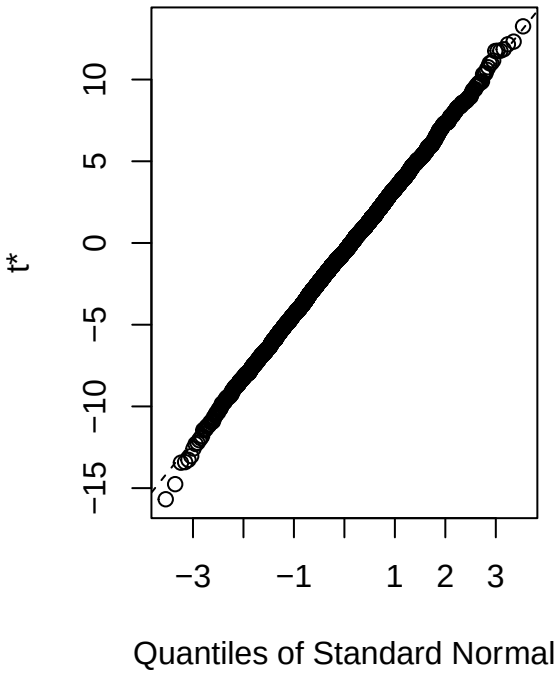
2.4.3) Model Diagnostics

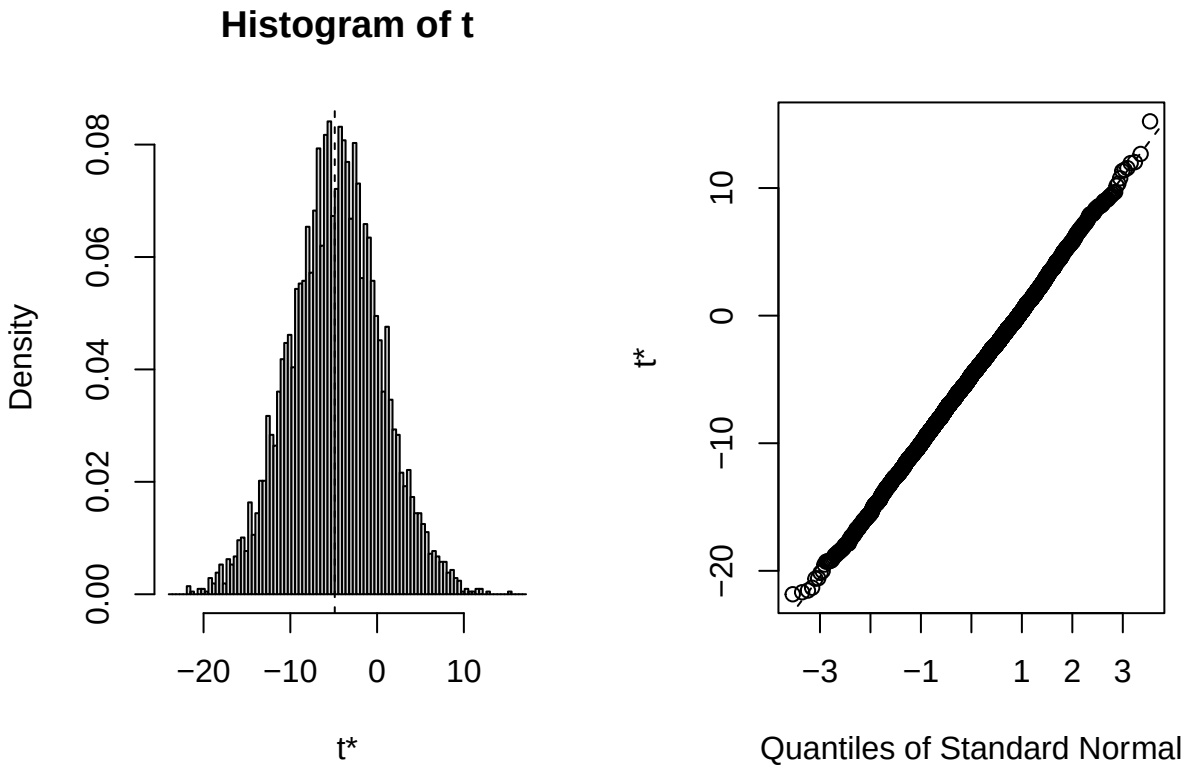

Histogram of t
Density
0.00
0.05
0.10
0.15
5
10
15
20
t*
t*
5
10
15
20
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0
1
2
3
4
5
0.2
0.4
0.6
0.8
t*
t*
0.3
0.4
0.5
0.6
0.7
0.8
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.04
0.08
−15
−5
0
5
10
t*
t*
−15
−10
−5
0
5
10
15
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.04
0.08
0.12
−15
−5
0
5
10
t*


t*
−15
−10
−5
0
5
10
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.02
0.04
0.06
0.08
−20
−10
0
10
t*
t*
10
0
−10
−20
−3
−1
1
2
3
Quantiles of Standard Normal

3) Child Prosocial at two weeks

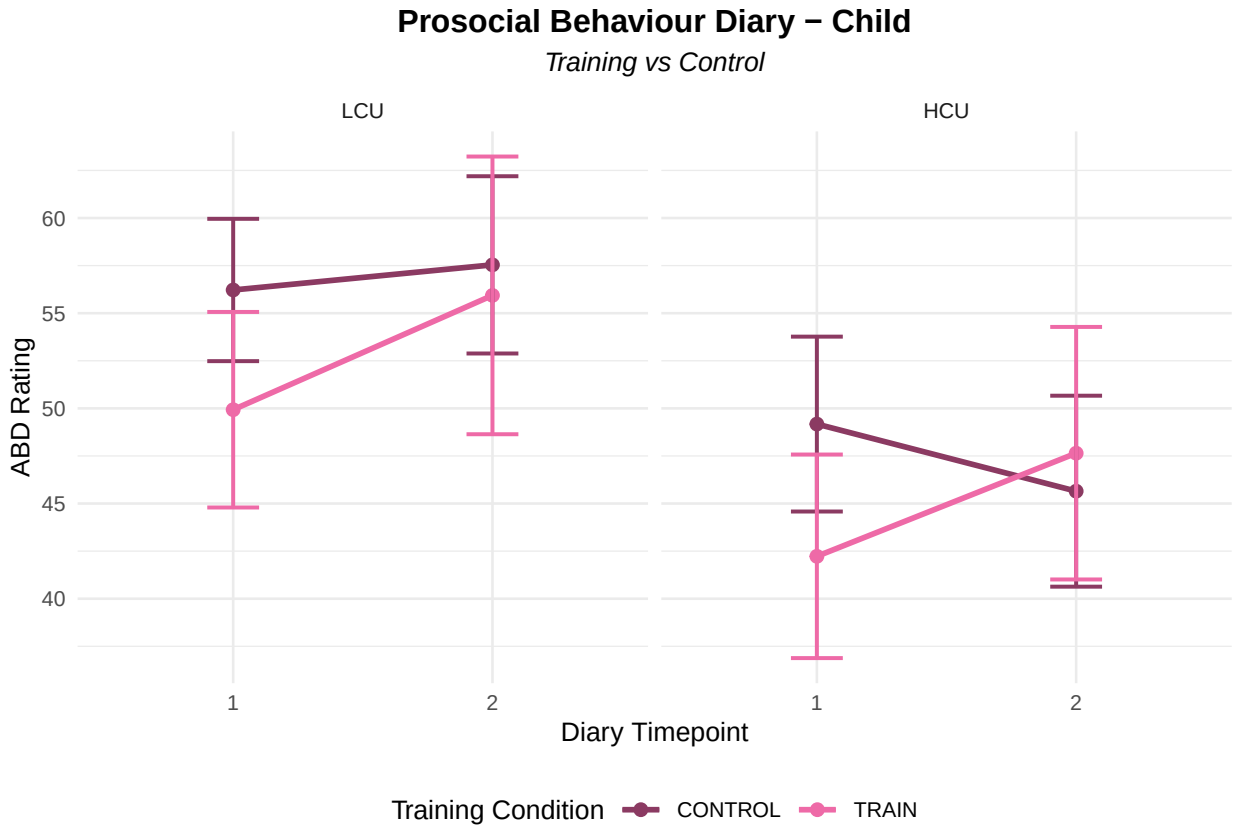
3.1) Table of outcomes

Table of outcomes:

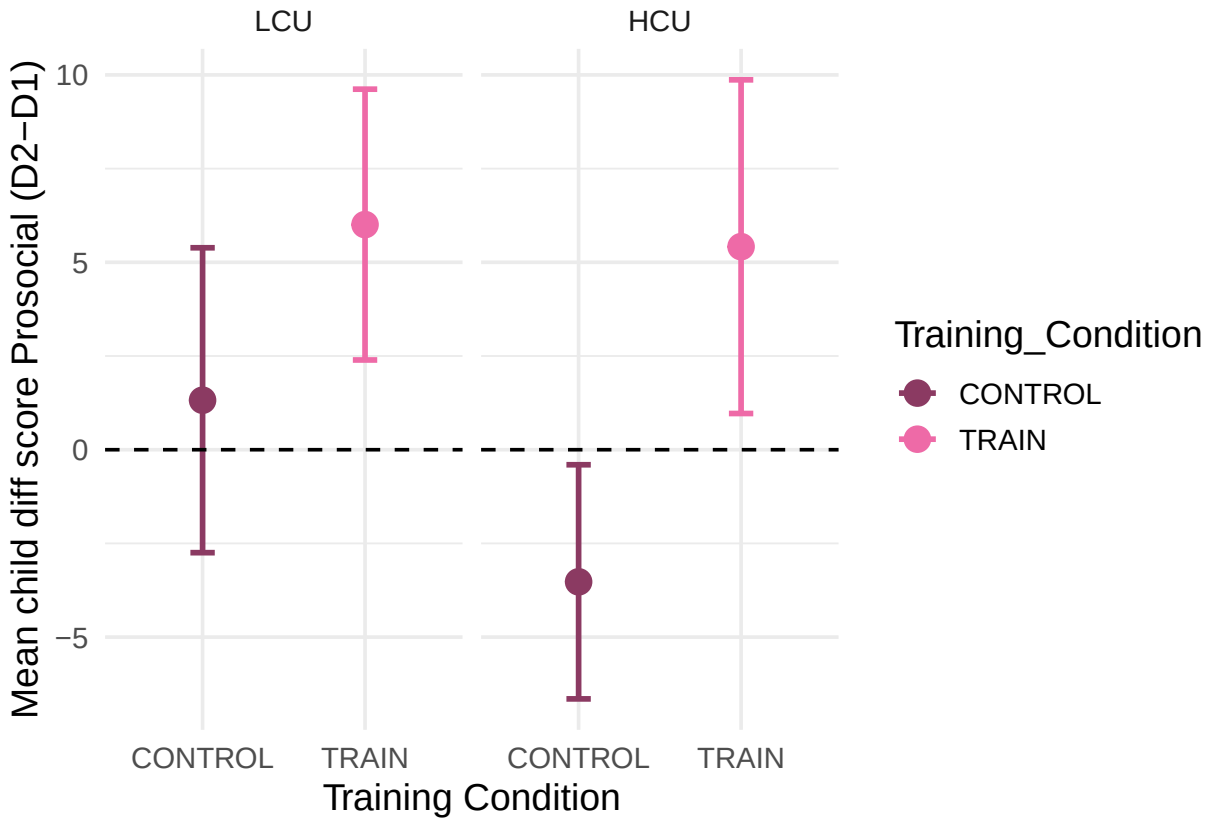
Training_Condition	diaryTP	CP_Group	Group_N_PBD	Mean_child_rated_PBD	se_child_rated_PBD	NA_child_rated_PBD
CONTROL	1	LCU	26	56.21923	3.738943	0
CONTROL	1	HCU	18	49.17444	4.593665	0
CONTROL	2	LCU	26	57.53923	4.655672	0
CONTROL	2	HCU	18	45.64944	5.016376	0
TRAIN	1	LCU	22	49.92909	5.136704	0
TRAIN	1	HCU	17	42.22706	5.347738	0
TRAIN	2	LCU	22	55.93545	7.295341	0
TRAIN	2	HCU	17	47.64471	6.633231	0

3.2) Plots of outcomes

First plotting data:



Difference plot:



3.3) Linear Regression

```
# singular fit so simplfying

model_c_psb_CU_lm <- lm(PSB_sum_D2 ~ PSB_sum_D1 + Training_Condition + CP_Group +
                          Training_Condition*CP_Group,
                        data =  child_psb_d1d2_wide)


# model without interaction
model_c_psb_CU_lm_nointeraction <- lm(PSB_sum_D2 ~ PSB_sum_D1 + Training_Condition + CP_Group,
                        data =  child_psb_d1d2_wide)
```

3.3.1) is model with interaction better fit? No

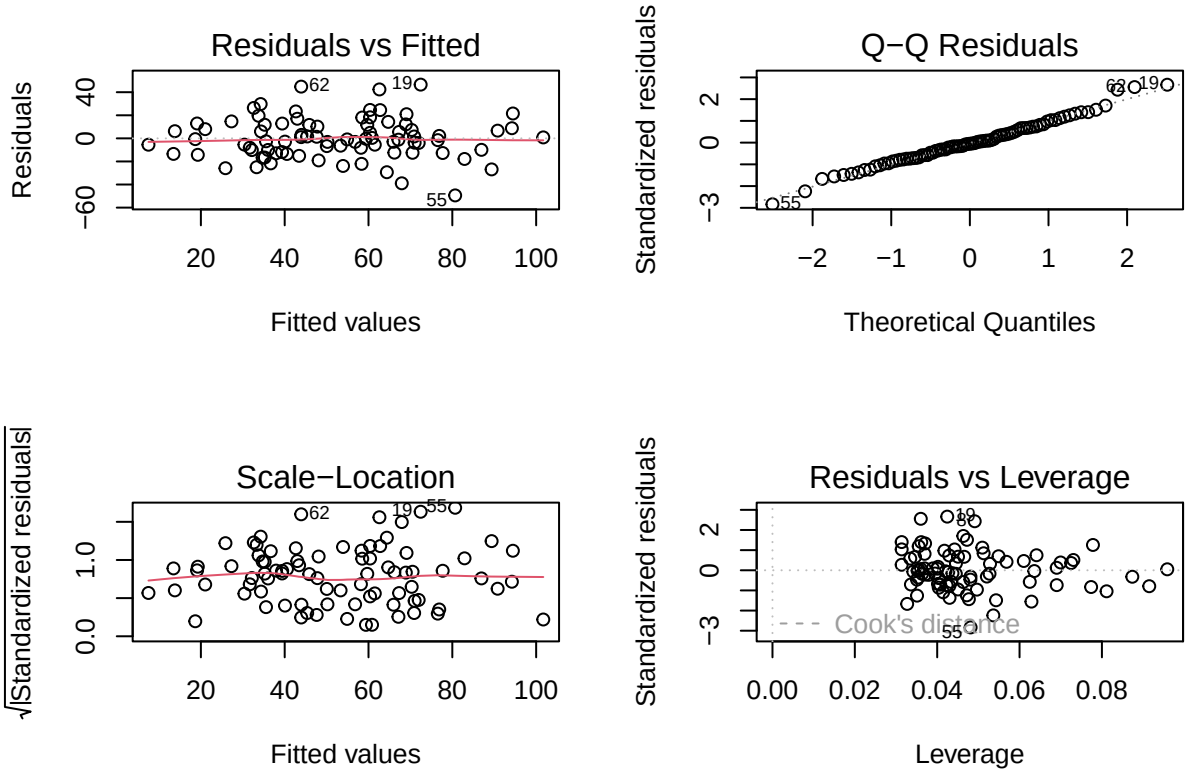
```
## Analysis of Variance Table
##
## Model 1: PSB_sum_D2 ~ PSB_sum_D1 + Training_Condition + CP_Group + Training_Condition *
##     CP_Group
## Model 2: PSB_sum_D2 ~ PSB_sum_D1 + Training_Condition + CP_Group
##   Res.Df   RSS Df Sum of Sq      F Pr(>F)
## 1     78 25123
## 2     79 25213 -1    -90.43 0.2808 0.5977
```

3.3.2) Model summary

```
##
## Call:
## lm(formula = PSB_sum_D2 ~ PSB_sum_D1 + Training_Condition + CP_Group,
##     data = child_psb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -49.437 -12.119  -0.904  11.662  46.556
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)                2.3638     6.1989   0.381    0.704
## PSB_sum_D1                 0.9668     0.0948  10.199 4.55e-16 ***
## Training_ConditionTRAIN    6.2697     3.9794   1.576    0.119
## CP_GroupHCU               -3.0710     4.0331  -0.761    0.449
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17.86 on 79 degrees of freedom
## Multiple R-squared:  0.5833, Adjusted R-squared:  0.5675
## F-statistic: 36.87 on 3 and 79 DF,  p-value: 5.29e-15


##
## Call:
## lm(formula = PSB_sum_D2 ~ PSB_sum_D1 + Training_Condition + CP_Group +
##     Training_Condition * CP_Group, data = child_psb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -50.268 -11.537   0.247  10.623  47.520
##
## Coefficients:
##                                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)                         3.16286    6.40729   0.494    0.623
## PSB_sum_D1                          0.96722    0.09523  10.156 6.33e-16 ***
## Training_ConditionTRAIN             4.48017    5.23327   0.856    0.395
## CP_GroupHCU                        -5.07593    5.54363  -0.916    0.363
## Training_ConditionTRAIN:CP_GroupHCU  4.23474    7.99201   0.530    0.598
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17.95 on 78 degrees of freedom
## Multiple R-squared:  0.5848, Adjusted R-squared:  0.5635
## F-statistic: 27.47 on 4 and 78 DF,  p-value: 3.075e-14
```

3.3.3) Assumption checks



3.4) Bootstrapped linear regression

3.4.1) Results

```
boot_coef_c_psb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(PSB_sum_D2 ~ PSB_sum_D1 + Training_Condition*CP_Group,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
## Boot function: Generates R bootstrap replicates of a statistic applied to data.
## data = my data
## statistic = A function which, when applied to data returns a vector containing the statistic(s) of i
set.seed(123)  # For reproducibility
boot_results_c_psb <- boot(data = child_psb_d1d2_wide,
                      statistic = boot_coef_c_psb,
                      R = 5000)

# View results
boot_results_c_psb
```

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = child_psb_d1d2_wide, statistic = boot_coef_c_psb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##       original       bias    std. error
## t1*  3.1628585  0.120389630  6.29606026
## t2*  0.9672201 -0.002332910  0.09302981
## t3*  4.4801737  0.059135088  5.48154848
## t4* -5.0759271 -0.049691120  5.13455240
## t5*  4.2347390 -0.006272314  7.56400773
```

3.4.2) Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_psb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   (-8.736, 16.514 )
## Calculations and Intervals on Original Scale
```

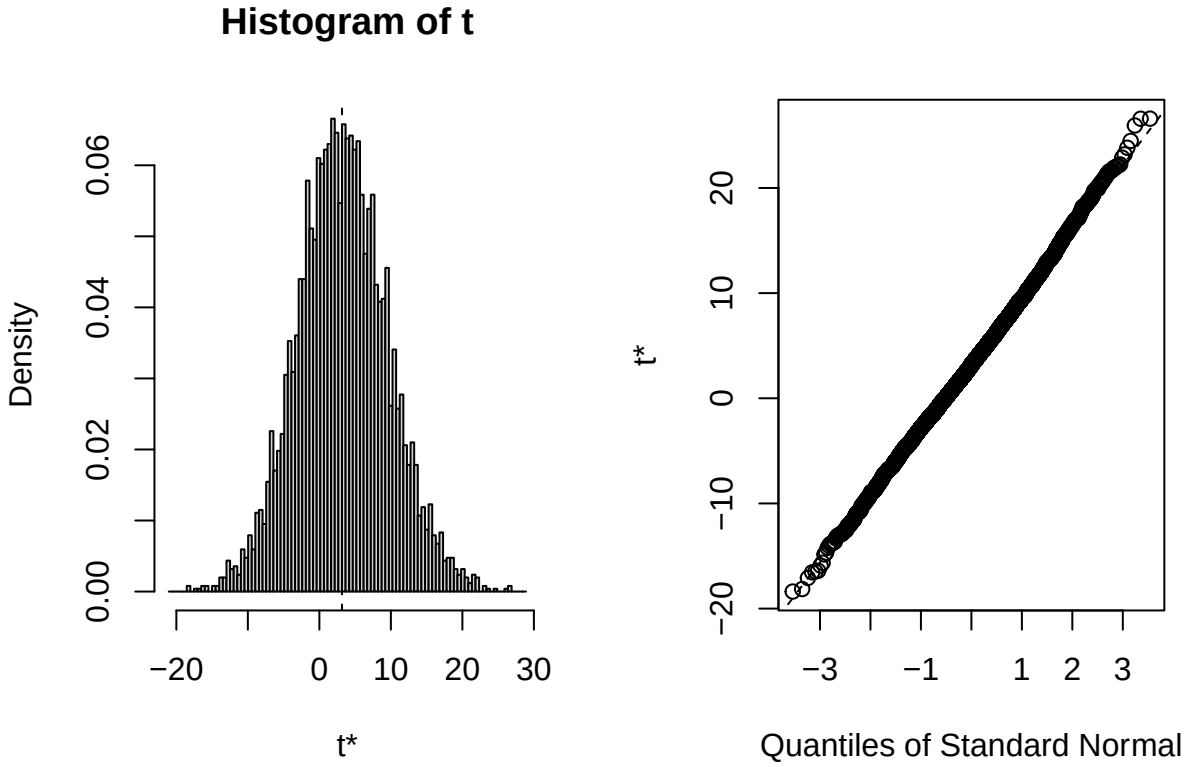
```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_psb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.7813,  1.1497 )
## Calculations and Intervals on Original Scale

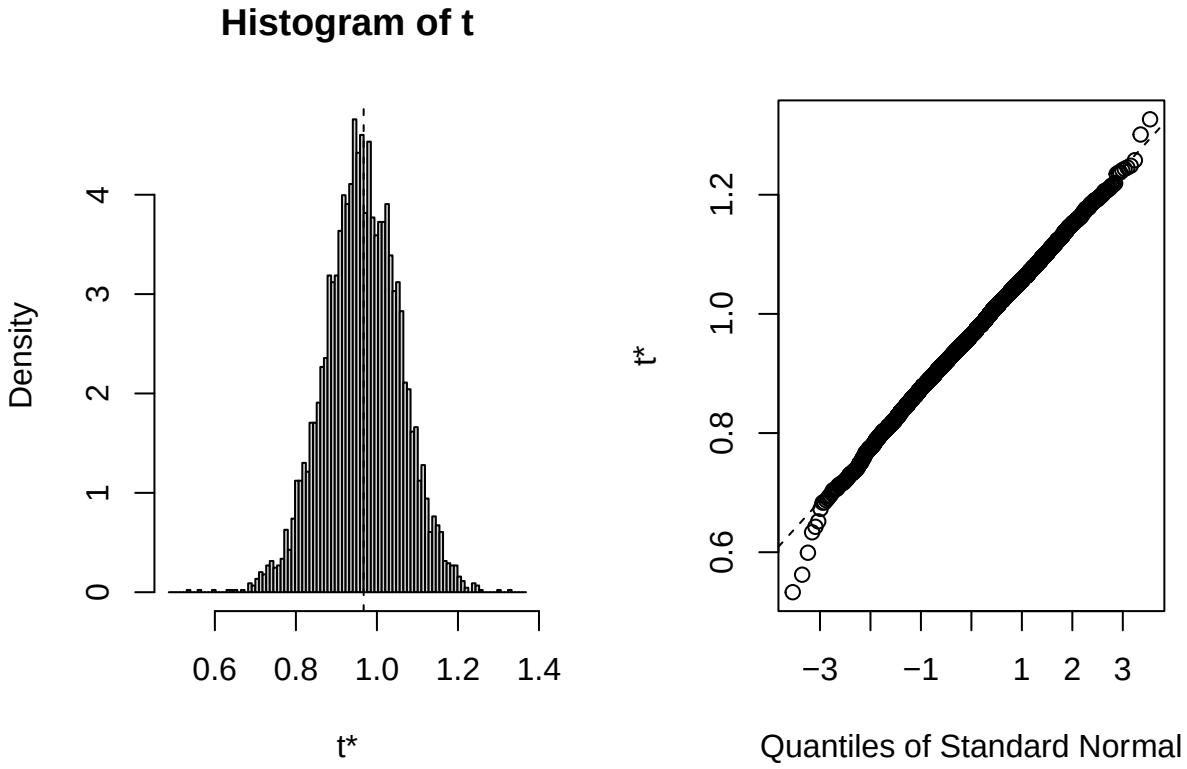
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_psb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-5.789, 15.692 )
## Calculations and Intervals on Original Scale

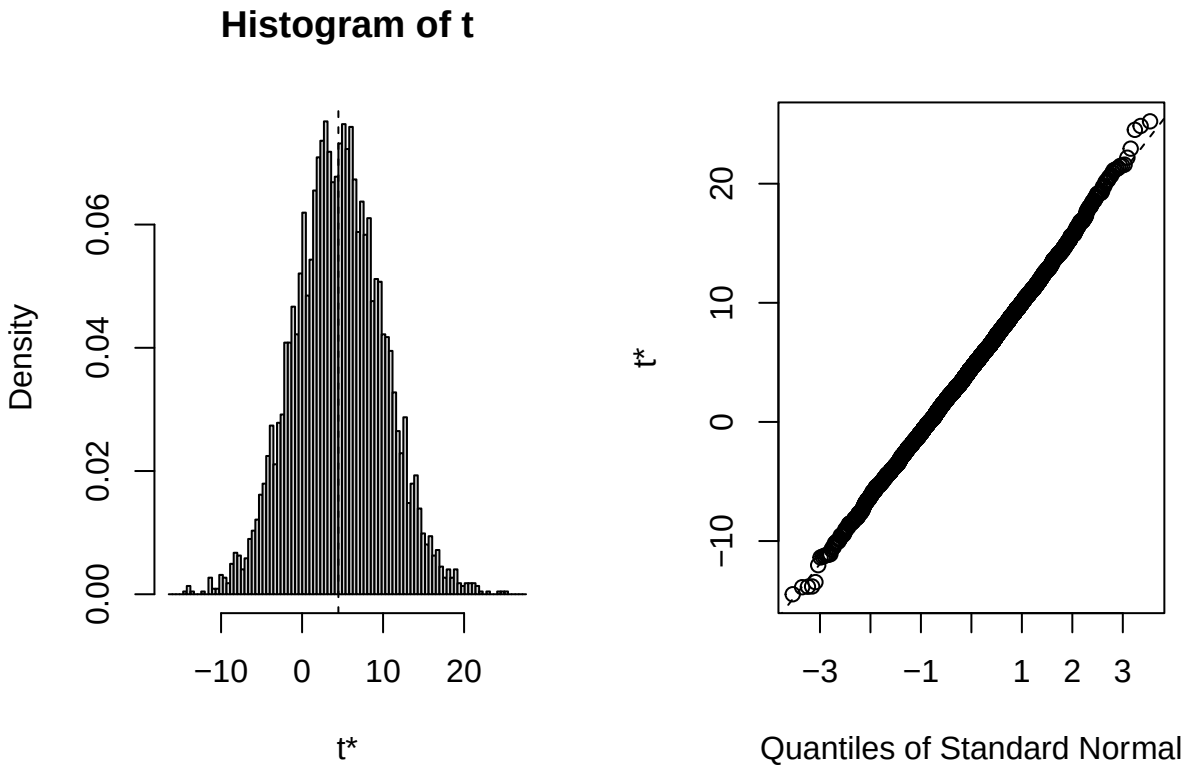
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_psb, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-15.800,   4.384 )
## Calculations and Intervals on Original Scale

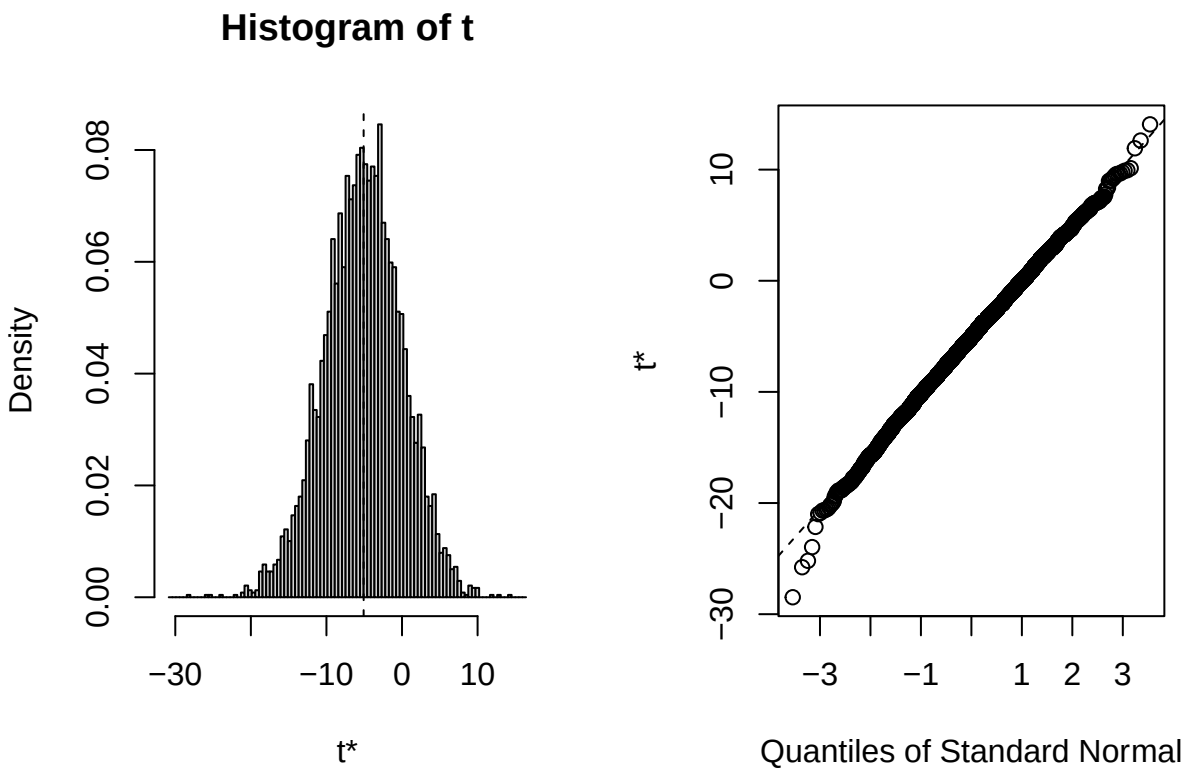
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_c_psb, type = "bca", index = 5)
##
## Intervals :
## Level       BCa
## 95%   (-9.892, 19.879 )
## Calculations and Intervals on Original Scale
```

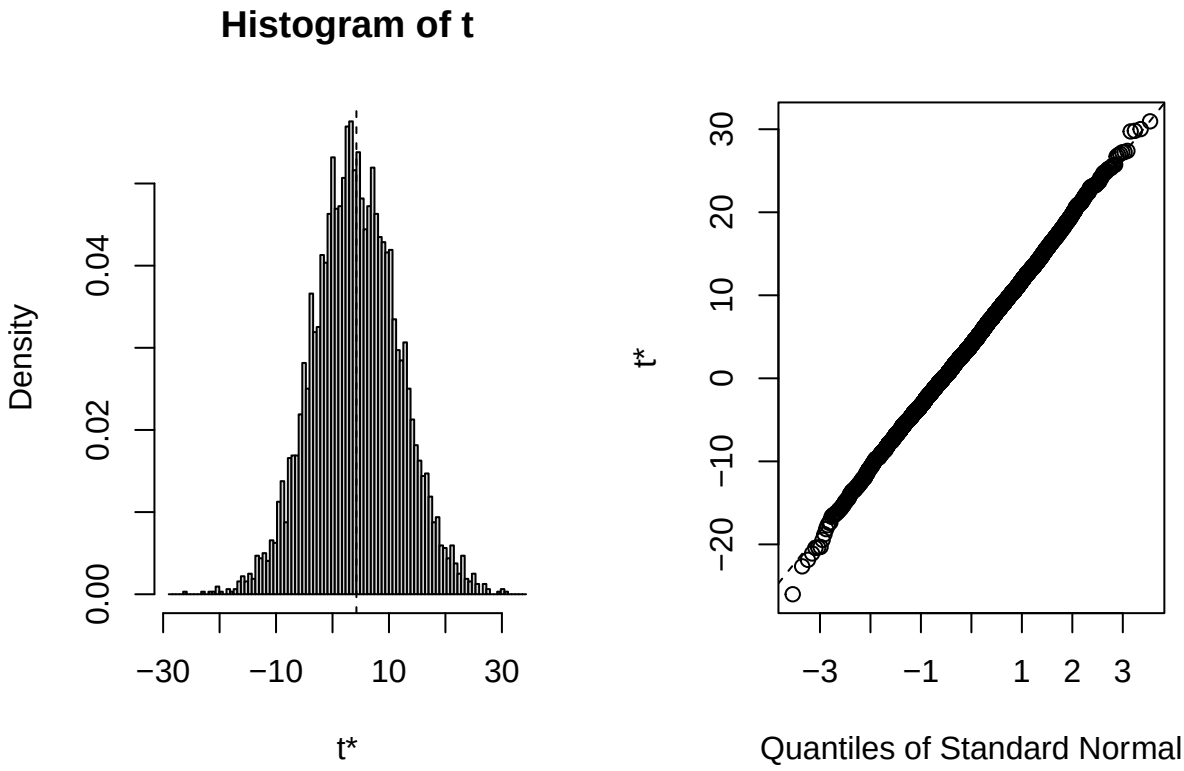
3.4.3) Model diagnostics






Histogram of t
Density
0.00
0.02
0.04
0.06
−10
0
10
20
t*
t*
−10
0
10
20
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.02
0.04
0.06
0.08
−30
−10
0
10
t*
t*
−30
−20
−10
0
10
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.02
0.04
−30
−10
10
30
t*
t*
30
20
10
0
−10
−20
−3
−1
1
2
3
Quantiles of Standard Normal

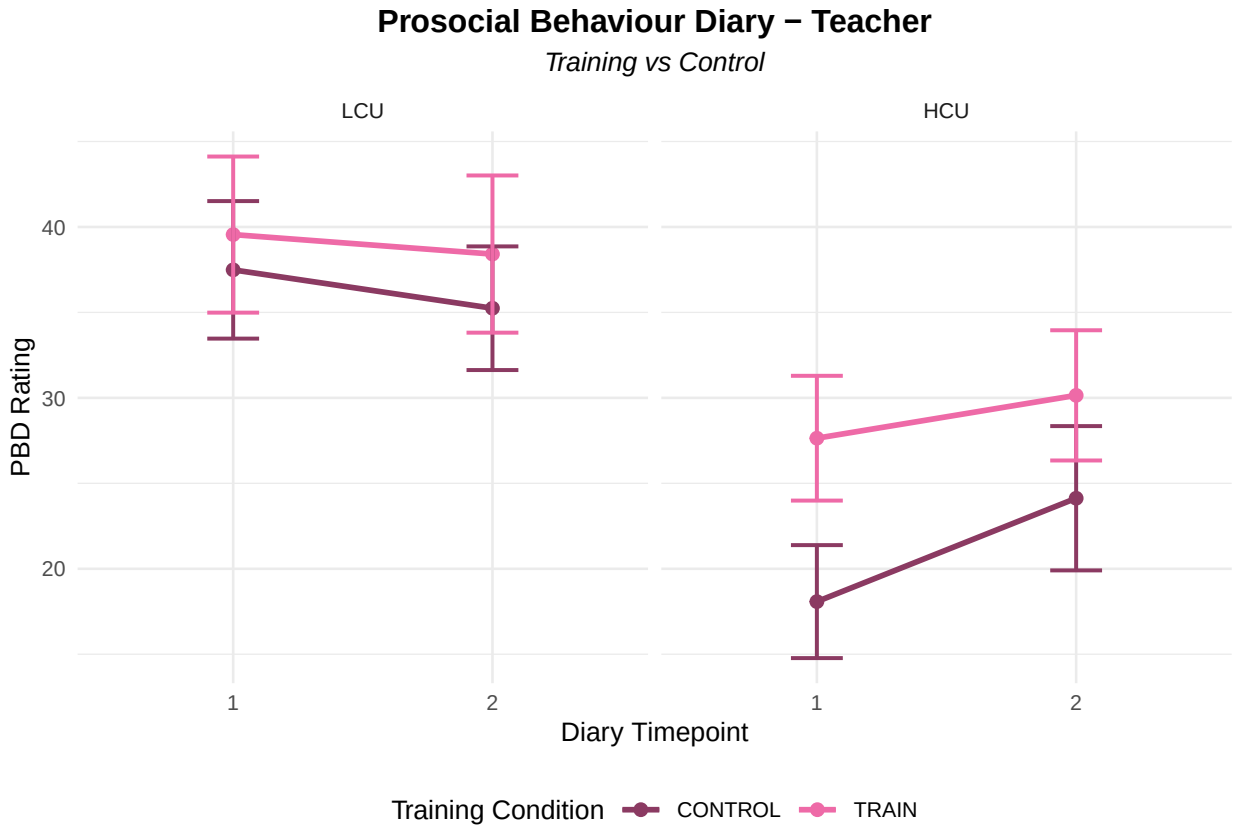
4) Teacher Prosocial at 2 weeks

4.1) Table of outcomes

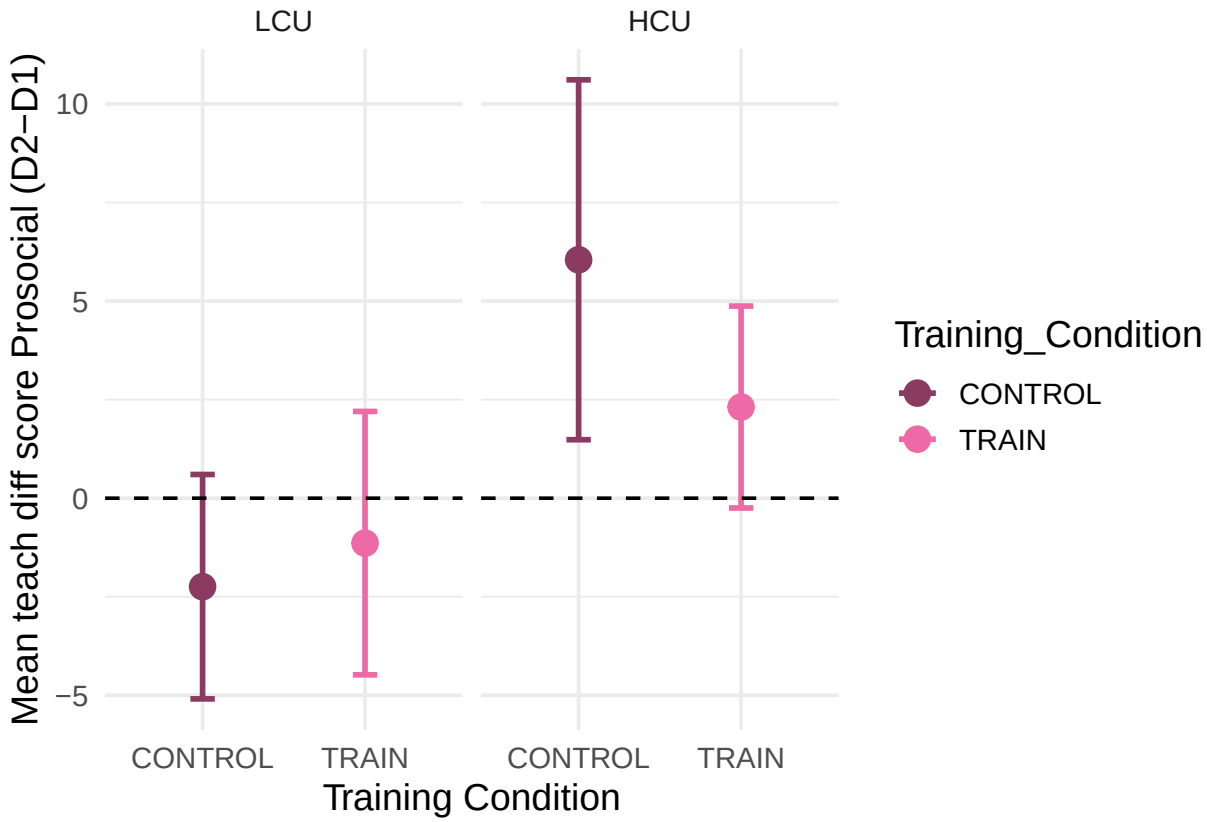
Training_Condition	diaryTP	CP_Group	Group_N	Mean_ABD	se_ABD	Mean_PBD	se_PBD
CONTROL	1	LCU	32	14.68229	3.379629	37.48958	4.023299
CONTROL	1	HCU	21	17.38889	3.301882	18.07937	3.305231
CONTROL	2	LCU	32	17.69792	3.158693	35.24479	3.616988
CONTROL	2	HCU	21	18.61905	3.537556	24.12698	4.220790
TRAIN	1	LCU	25	21.24000	4.115801	39.55333	4.566927
TRAIN	1	HCU	25	21.40000	3.943220	27.64000	3.650338
TRAIN	2	LCU	25	22.10000	2.898211	38.41333	4.598401
TRAIN	2	HCU	25	16.68000	3.256161	30.14667	3.809410

4.2) Plots of outcomes

First plotting data:



Difference plot:


LCU
HCU
Mean teach diff score Prosocial (D2−D1)
10
5
0
−5
CONTROL
TRAIN
CONTROL
TRAIN
Training Condition
Training_Condition
CONTROL
TRAIN

4.3) Linear Regression model

```
model_t_psb_CU_lm <- lm(PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition + CP_Group +
                         Training_Condition*CP_Group,
                       data =  teach_psb_d1d2_wide)

# model without interaction
model_t_psb_CU_lm_nointeraction <- lm(PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition + CP_Group,
                       data =  teach_psb_d1d2_wide)
```

4.3.1) Does interaction term improve model? No

```
## Analysis of Variance Table
##
## Model 1: PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition + CP_Group + Training_Condition *
##     CP_Group
## Model 2: PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition + CP_Group
##   Res.Df   RSS Df Sum of Sq     F Pr(>F)
## 1     97 22866
## 2     98 22892 -1   -26.406 0.112 0.7386
```

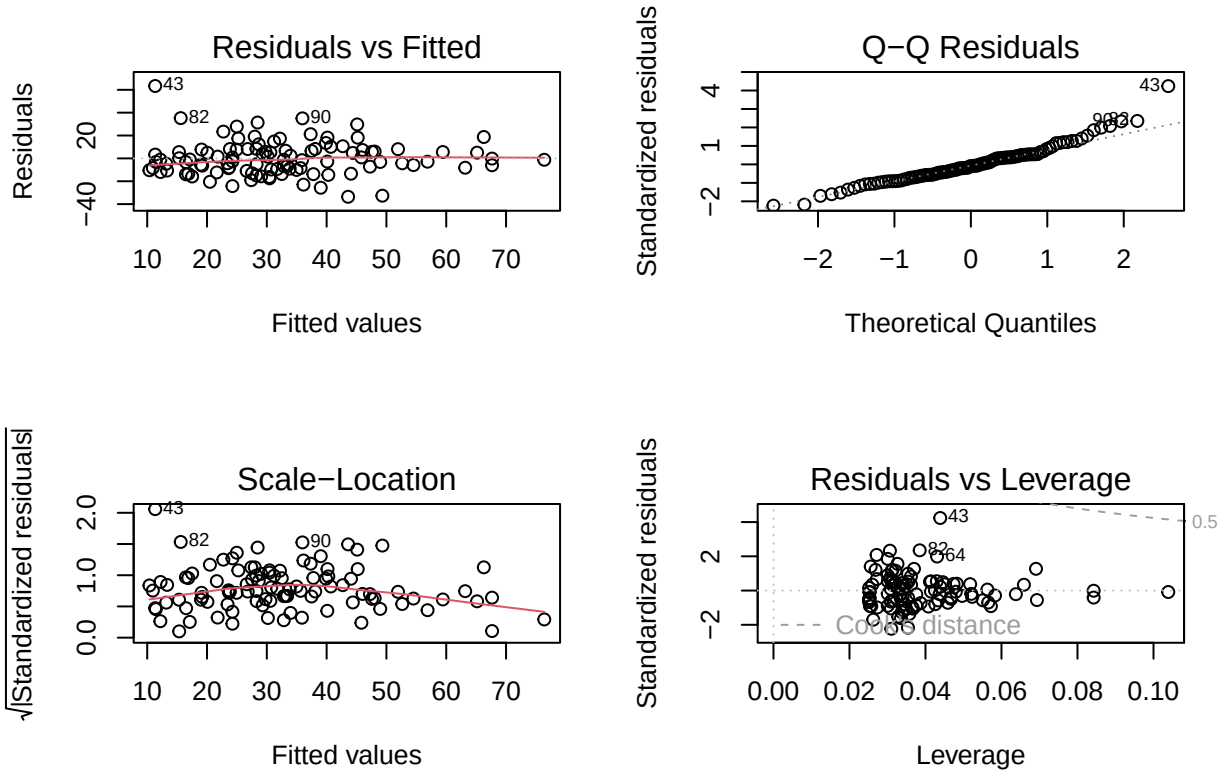
4.3.2) Model results

```
##
## Call:
## lm(formula = PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition +
##     CP_Group, data = teach_psb_d1d2_wide)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -33.596  -9.616  -1.385   7.618  63.332
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)              10.37655    3.63357   2.856  0.00524 **
## PSB_mean_D1               0.67397    0.07555   8.921 2.65e-14 ***
## Training_ConditionTRAIN  0.86896    3.07457   0.283  0.77806
## CP_GroupHCU               0.95821    3.26973   0.293  0.77010
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.28 on 98 degrees of freedom
## Multiple R-squared:  0.4801, Adjusted R-squared:  0.4642
## F-statistic: 30.16 on 3 and 98 DF,  p-value: 6.712e-14


##
## Call:
## lm(formula = PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition +
##     CP_Group + Training_Condition * CP_Group, data = teach_psb_d1d2_wide)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -34.129  -9.414  -1.519   7.234  62.773
##
## Coefficients:
##                                    Estimate Std. Error t value Pr(>|t|)
## (Intercept)                         9.87682    3.94372   2.504   0.0139 *
## PSB_mean_D1                         0.67667    0.07632   8.866 3.75e-14 ***
## Training_ConditionTRAIN             1.77207    4.10127   0.432   0.6666
## CP_GroupHCU                         2.01645    4.55918   0.442   0.6593
## Training_ConditionTRAIN:CP_GroupHCU -2.07049    6.18627  -0.335   0.7386
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.35 on 97 degrees of freedom
## Multiple R-squared:  0.4807, Adjusted R-squared:  0.4593
## F-statistic: 22.45 on 4 and 97 DF,  p-value: 3.839e-13
```

4.3.3) Assumption checks



4.4) Bootstrapped model

4.4.1) Results and specification

```
boot_coef_t_psb <- function(data, indices) {
  d <- data[indices, ]  # Resample with replacement
  fit <- lm(PSB_mean_D2 ~ PSB_mean_D1 + Training_Condition*CP_Group,
                    data =  d)
  return(coef(fit))
}

# Run bootstrap
## Boot function: Generates R bootstrap replicates of a statistic applied to data.
## data = my data
## statistic = A function which, when applied to data returns a vector containing the statistic(s) of i
set.seed(14423)  # For reproducibility
boot_results_t_psb <- boot(data = teach_psb_d1d2_wide,
                     statistic = boot_coef_t_psb,
                     R = 5000)

# View results
boot_results_t_psb
```

```
##
## ORDINARY NONPARAMETRIC BOOTSTRAP
##
##
## Call:
## boot(data = teach_psb_d1d2_wide, statistic = boot_coef_t_psb,
##     R = 5000)
##
##
## Bootstrap Statistics :
##       original        bias    std. error
## t1*  9.8768193  0.0917269268   3.6543584
## t2*  0.6766672 -0.0008235652   0.0617698
## t3*  1.7720697 -0.0663220366   3.9550263
## t4*  2.0164510 -0.1739272737   4.6541320
## t5* -2.0704949  0.2548293215   6.2384329
```

4.4.2) Confidence Intervals

```
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, type = "bca", index = 1)
##
## Intervals :
## Level       BCa
## 95%   ( 3.333, 17.668 )
## Calculations and Intervals on Original Scale

## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, type = "bca", index = 2)
##
## Intervals :
## Level       BCa
## 95%   ( 0.5420,  0.7895 )
## Calculations and Intervals on Original Scale

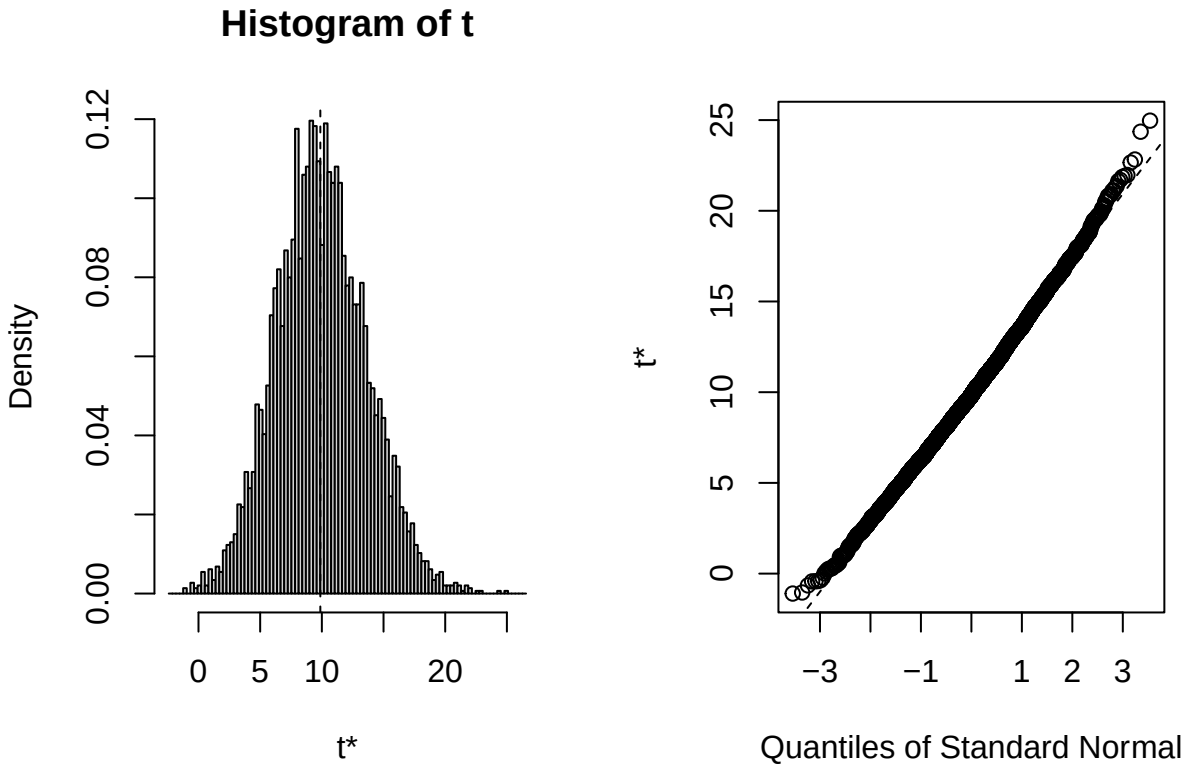
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, type = "bca", index = 3)
##
## Intervals :
## Level       BCa
## 95%   (-6.200,  8.997 )
## Calculations and Intervals on Original Scale

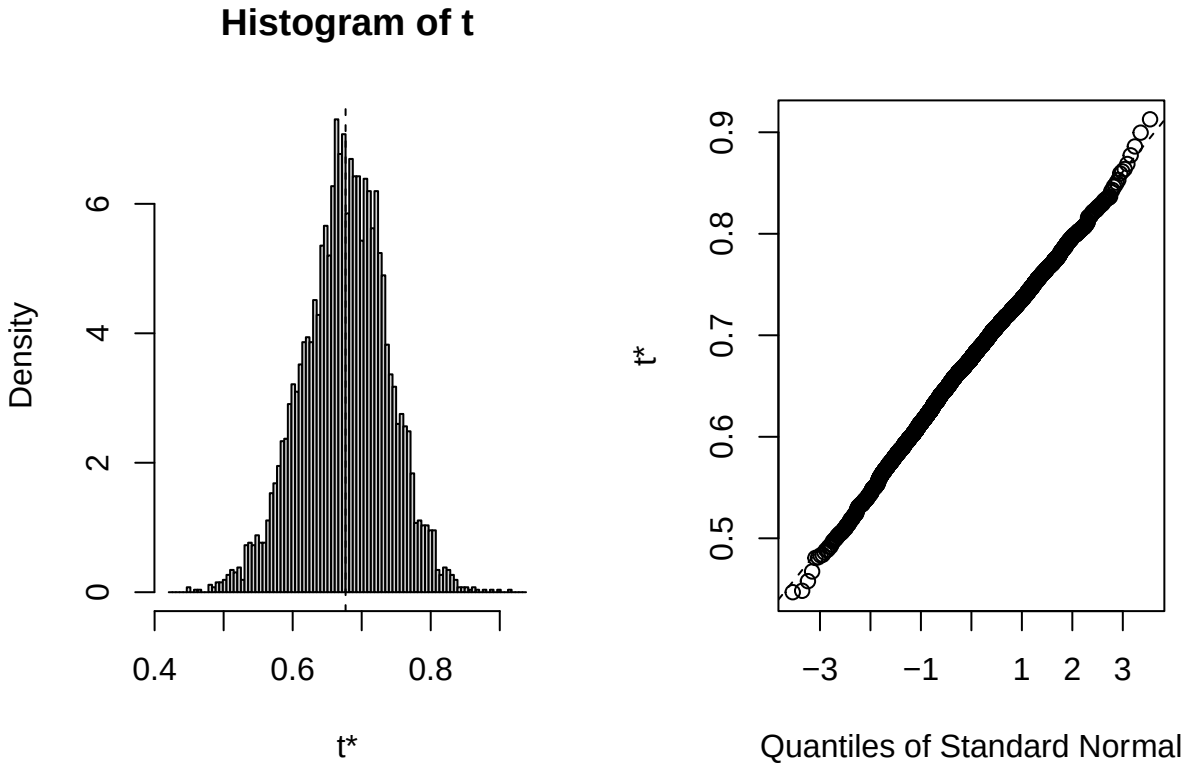
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
```

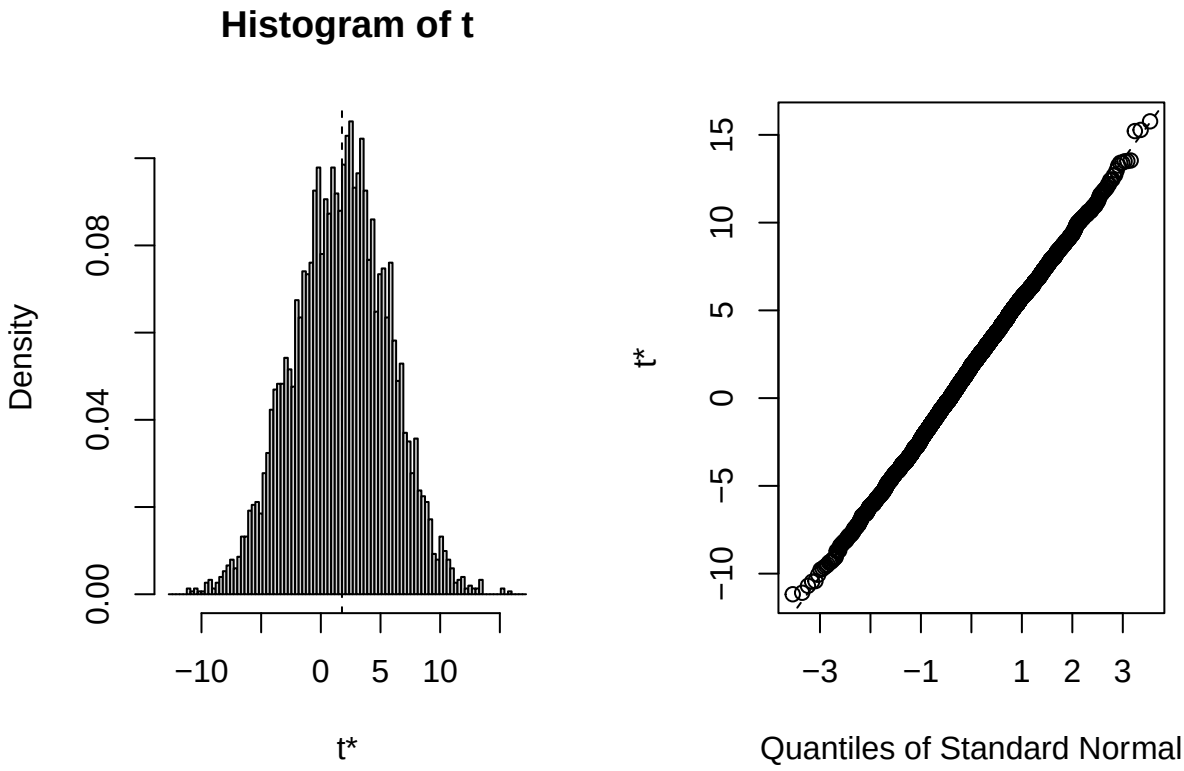
```
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, type = "bca", index = 4)
##
## Intervals :
## Level       BCa
## 95%   (-6.273, 12.297 )
## Calculations and Intervals on Original Scale

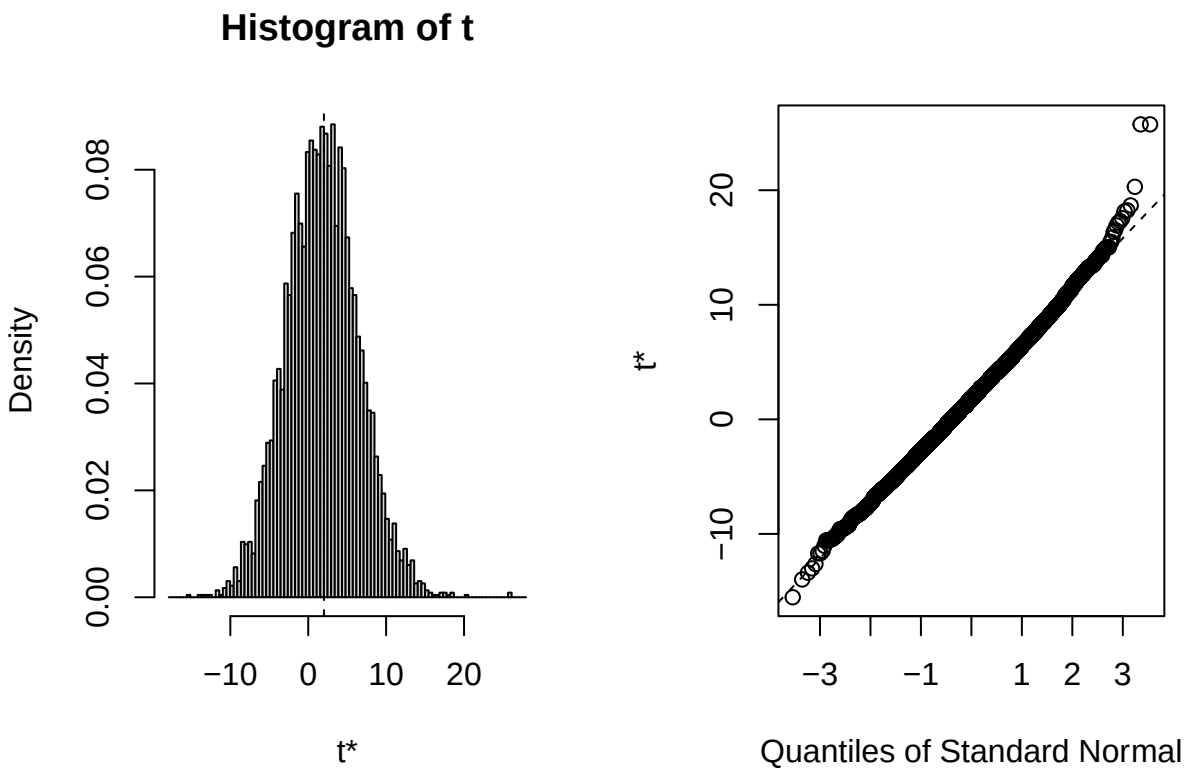
## BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
## Based on 5000 bootstrap replicates
##
## CALL :
## boot.ci(boot.out = boot_results_t_psb, type = "bca", index = 5)
##
## Intervals :
## Level       BCa
## 95%   (-14.596,   9.445 )
## Calculations and Intervals on Original Scale
```

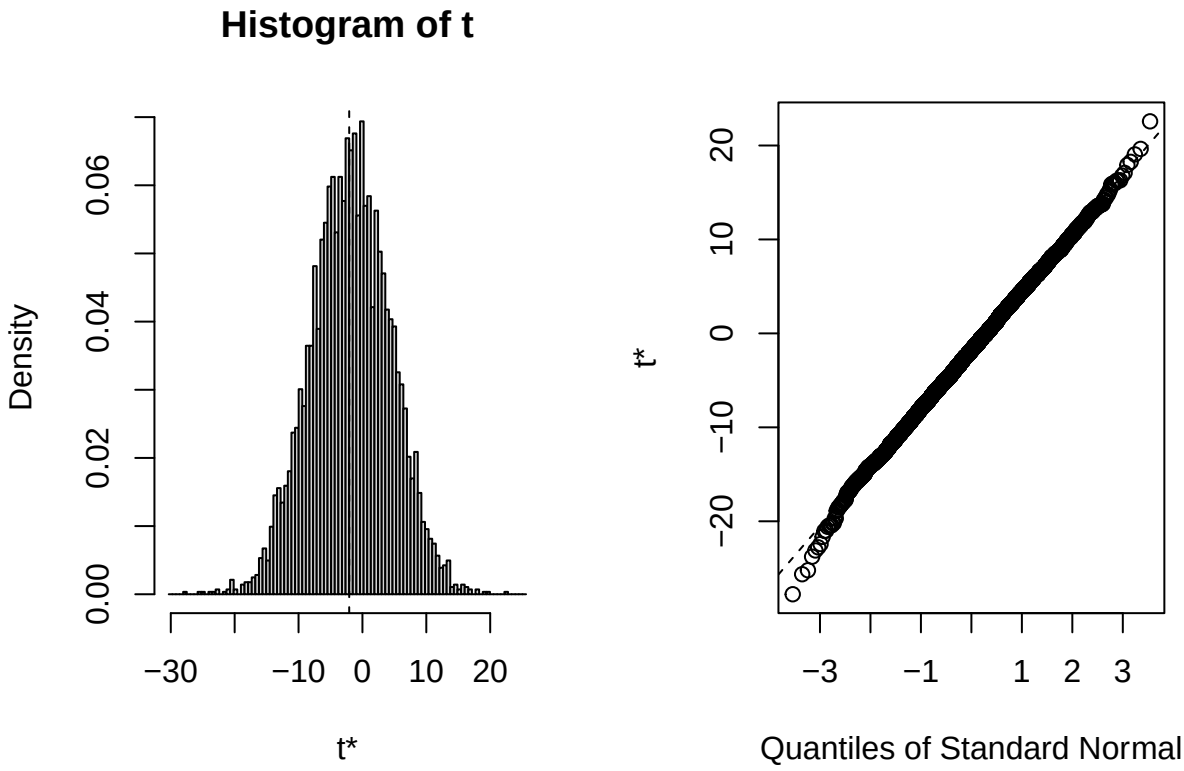
4.4.5) Model diagnostics






Histogram of t
Density
0.00
0.04
0.08
−10
0
5
10
t*
t*
−10
−5
0
5
10
15
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.02
0.04
0.06
0.08
−10
0
10
20
t*
t*
−10
0
10
20
−3
−1
1
2
3
Quantiles of Standard Normal


Histogram of t
Density
0.00
0.02
0.04
0.06
−30
−10
0
10
20
t*
t*
−20
−10
0
10
20
−3
−1
1
2
3
Quantiles of Standard Normal